

them from the horizontal planar connections along the interconnect layers of the IC dies.

26 Claims, 27 Drawing Sheets

Related U.S. Application Data

continuation of application No. 16/159,705, filed on Oct. 14, 2018, now Pat. No. 10,672,745, which is a continuation-in-part of application No. 15/976,809, filed on May 10, 2018, now Pat. No. 10,580,735, said application No. 16/833,341 is a continuation-in-part of application No. 15/859,546, filed on Dec. 31, 2017, now Pat. No. 10,607,136, said application No. 16/159,705 is a continuation-in-part of application No. 15/859,546, filed on Dec. 31, 2017, now Pat. No. 10,607,136, and a continuation-in-part of application No. 15/859,551, filed on Dec. 31, 2017, now Pat. No. 11,176,450, and a continuation-in-part of application No. 15/859,548, filed on Dec. 31, 2017, now Pat. No. 10,719,762, and a continuation-in-part of application No. 15/859,612, filed on Dec. 31, 2017, now Pat. No. 10,762,420, said application No. 15/976,809 is a continuation-in-part of application No. 15/725,030, filed on Oct. 4, 2017, now Pat. No. 10,522,352, said application No. 16/159,705 is a continuation-in-part of application No. 15/725,030, filed on Oct. 4, 2017, now Pat. No. 10,522,352.

- (60) Provisional application No. 62/678,246, filed on May 30, 2018, provisional application No. 62/619,910, filed on Jan. 21, 2018, provisional application No. 62/575,221, filed on Oct. 20, 2017, provisional application No. 62/575,259, filed on Oct. 20, 2017, provisional application No. 62/575,184, filed on Oct. 20, 2017, provisional application No. 62/575,240, filed on Oct. 20, 2017, provisional application No. 62/541,064, filed on Aug. 3, 2017, provisional application No. 62/405,833, filed on Oct. 7, 2016.

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H01L 25/18 (2023.01)
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H01L 25/07 (2006.01)
H01L 25/075 (2006.01)
H01L 25/10 (2006.01)
H01L 25/11 (2006.01)
H01L 25/16 (2023.01)
H10B 51/20 (2023.01)
H10B 53/20 (2023.01)
H10B 63/00 (2023.01)
H10D 88/00 (2025.01)
H10K 19/00 (2023.01)

(52) U.S. Cl.

CPC *H01L 24/03* (2013.01); *H01L 24/08* (2013.01); *H01L 24/11* (2013.01); *H01L 24/16* (2013.01); *H01L 24/32* (2013.01); *H01L 24/80* (2013.01); *H01L 25/043* (2013.01); *H01L 25/074* (2013.01); *H01L 25/0756* (2013.01); *H01L 25/105* (2013.01); *H01L 25/117* (2013.01); *H01L 25/16* (2013.01); *H01L 2224/08145* (2013.01); *H01L 2224/16145* (2013.01); *H01L 2224/16225* (2013.01); *H01L*

2224/17181 (2013.01); *H01L 2224/32145* (2013.01); *H01L 2224/80895* (2013.01); *H01L 2224/80896* (2013.01); *H01L 2224/9202* (2013.01); *H01L 2225/06503* (2013.01); *H01L 2225/06548* (2013.01); *H10B 41/20* (2023.02); *H10B 51/20* (2023.02); *H10B 53/20* (2023.02); *H10B 63/84* (2023.02); *H10D 88/00* (2025.01); *H10K 19/201* (2023.02)

(58) Field of Classification Search

CPC *H01L 25/117*; *H01L 25/16*; *H01L 25/18*; *H01L 2225/06503*; *H01L 2225/06548*; *H01L 2225/06524*; *H01L 2225/06565*; *H01L 2225/06541*; *H01L 2224/08145*; *H01L 2224/16145*; *H01L 2224/16225*; *H01L 2224/17181*; *H01L 2224/32145*; *H01L 27/0688*; *H01L 27/281*; *H01L 27/11514*; *H01L 27/11551*; *H01L 27/11597*; *H01L 27/2481*; *H01L 24/08*; *H01L 24/16*; *H01L 24/32*; *H10B 41/20*; *H10B 51/20*; *H10B 53/20*; *H10K 19/201*; *H10D 88/00*

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

5,376,825 A	12/1994	Tukamoto et al.	
5,579,207 A	11/1996	Hayden et al.	
5,621,863 A	4/1997	Boulet et al.	
5,673,478 A	10/1997	Beene et al.	
5,717,832 A	2/1998	Steimle et al.	
5,740,326 A	4/1998	Boulet et al.	
5,793,115 A	8/1998	Zavracky	
5,909,587 A	6/1999	Tran	
6,320,255 B1	11/2001	Terrill	
6,421,654 B1	7/2002	Gordon	
6,624,046 B1 *	9/2003	Zavracky	<i>H01L 23/49833</i>
			257/E21.705
6,642,081 B1 *	11/2003	Patti	<i>H01L 23/481</i>
			257/E21.597
6,707,124 B2	3/2004	Wachtler et al.	
6,844,624 B1	1/2005	Kiritani	
6,891,447 B2	5/2005	Song	
6,909,194 B2	6/2005	Farnworth et al.	
6,917,219 B2	7/2005	New	
6,962,835 B2	11/2005	Tong et al.	
7,046,522 B2	5/2006	Sung et al.	
7,099,215 B1	8/2006	Rotenberg et al.	
7,124,250 B2	10/2006	Kyung	
7,202,566 B2	4/2007	Liaw	
7,485,968 B2	2/2009	Enquist et al.	
7,638,869 B2	12/2009	Irsigler et al.	
7,692,946 B2	4/2010	Taufique et al.	
7,863,918 B2	1/2011	Jenkins	
7,902,851 B2 *	3/2011	Fenner	<i>H05K 1/11</i>
			324/762.02
7,977,962 B2	7/2011	Hargan et al.	
8,032,711 B2	10/2011	Black et al.	
8,042,082 B2	10/2011	Solomon	
8,053,277 B2 *	11/2011	Yu	<i>H01L 27/0688</i>
			257/E21.511
8,059,443 B2	11/2011	McLaren et al.	
8,110,899 B2	2/2012	Reed et al.	
8,148,814 B2	4/2012	Furuta et al.	
8,198,716 B2 *	6/2012	Periaman	<i>H01L 25/18</i>
			257/E21.597
8,223,523 B2	7/2012	Jin et al.	
8,223,524 B2	7/2012	Chung	
8,228,684 B2	7/2012	Losavio et al.	
8,233,303 B2	7/2012	Best et al.	
8,432,467 B2	4/2013	Jaworski et al.	
8,441,831 B2	5/2013	Ku et al.	
8,516,409 B2	8/2013	Coteus et al.	

(56)

References Cited

U.S. PATENT DOCUMENTS

8,546,188 B2 *	10/2013	Liu	H01L 23/481	10,600,780 B2	3/2020	DeLaCruz et al.
			438/455	10,607,136 B2	3/2020	Teig et al.
8,546,955 B1	10/2013	Wu		10,672,663 B2	6/2020	DeLaCruz et al.
8,547,769 B2	10/2013	Saraswat et al.		10,672,743 B2	6/2020	Teig et al.
8,552,569 B2	10/2013	Lee		10,672,744 B2	6/2020	Teig et al.
8,697,493 B2 *	4/2014	Sadaka	H01L 24/80	10,672,745 B2	6/2020	Teig et al.
			438/455	10,719,762 B2	7/2020	Teig et al.
8,704,384 B2	4/2014	Wu et al.		10,762,420 B2	9/2020	Teig et al.
8,736,068 B2	5/2014	Bartley et al.		2001/0017418 A1	8/2001	Noguchi et al.
8,797,818 B2	8/2014	Jeddeloh		2002/0008309 A1	1/2002	Akiyama
8,802,541 B2 *	8/2014	Wang	H01L 24/80	2003/0102495 A1	6/2003	Huppenthal et al.
			438/452	2004/0178819 A1 *	9/2004	New H01L 23/3171
8,816,486 B2 *	8/2014	Tsai	H01L 24/05			257/E27.11
			257/777	2005/0127490 A1	6/2005	Black et al.
8,816,506 B2	8/2014	Kawashita et al.		2006/0036559 A1	2/2006	Nugent
8,860,199 B2	10/2014	Black et al.		2006/0087013 A1	4/2006	Hsieh
8,860,221 B2 *	10/2014	Park	H01L 24/80	2006/0097402 A1 *	5/2006	Pu H01L 25/03
			257/E23.161			257/784
8,885,380 B2	11/2014	Kang et al.		2007/0220207 A1	9/2007	Black et al.
8,901,749 B2	12/2014	Kim et al.		2008/0150088 A1 *	6/2008	Reed H01L 25/0652
8,907,439 B1	12/2014	Kay et al.		2008/0225493 A1 *	9/2008	Barth H01L 25/0657
8,930,647 B1	1/2015	Smith				361/717
8,947,931 B1	2/2015	D'Abreu		2008/0237310 A1	10/2008	Periaman et al.
8,987,066 B2	3/2015	Grobelny et al.		2008/0237891 A1 *	10/2008	Irsigler H01L 24/73
9,030,253 B1	5/2015	Ngai				257/778
9,064,937 B2 *	6/2015	Edelstein	H01L 23/53233	2009/0037658 A1 *	2/2009	Sistla G06F 12/084
9,067,272 B2	6/2015	Sutanto				711/119
9,076,700 B2	7/2015	Kawashita et al.		2009/0070727 A1	3/2009	Solomon
9,076,770 B2	7/2015	Foster, Sr. et al.		2009/0138688 A1 *	5/2009	Black H01L 25/0657
9,142,262 B2	9/2015	Ware				712/E9.045
9,224,704 B2 *	12/2015	Landru	H01L 25/50	2009/0224388 A1 *	9/2009	Bernstein H01L 25/0652
9,230,940 B2	1/2016	Goodnow et al.				438/109
9,300,298 B2	3/2016	Cordero		2010/0140750 A1	6/2010	Toms
9,318,418 B2	4/2016	Kawashita et al.		2010/0255262 A1 *	10/2010	Chen B23K 20/233
9,418,964 B2	8/2016	Chang et al.				228/164
9,432,298 B1	8/2016	Smith		2010/0258890 A1 *	10/2010	Ahn H01L 24/81
9,478,496 B1 *	10/2016	Lin	H01L 24/83			257/E31.127
9,484,326 B2	11/2016	Keeth et al.		2010/0261159 A1	10/2010	Hess et al.
9,497,854 B2	11/2016	Giuliano		2010/0308455 A1 *	12/2010	Kim H01L 24/91
9,501,603 B2	11/2016	Barowski et al.				257/734
9,508,607 B2	11/2016	Chua-Eoan		2011/0026293 A1	2/2011	Riho
9,559,081 B1 *	1/2017	Lai	H01L 23/5389	2011/0078412 A1 *	3/2011	Emma G06F 9/3885
9,613,925 B2 *	4/2017	Cai	H01L 24/14			712/30
9,640,233 B2	5/2017	Sohn		2011/0102066 A1 *	5/2011	Jin G11C 5/04
9,645,603 B1	5/2017	Chall et al.				327/524
9,647,187 B1 *	5/2017	Yap	G01S 7/4817	2011/0131391 A1	6/2011	Barowski et al.
9,679,867 B2 *	6/2017	Ashidate	H01L 24/80	2011/0248396 A1 *	10/2011	Liu H01L 25/0657
9,691,739 B2	6/2017	Kawashita et al.				438/459
9,726,691 B2	8/2017	Garibay		2012/0092062 A1	4/2012	Lee et al.
9,746,517 B2	8/2017	Whetsel		2012/0098140 A1 *	4/2012	Bartley H01L 25/18
9,747,959 B2	8/2017	Seo				257/E23.174
9,871,014 B2	1/2018	Haba		2012/0119357 A1	5/2012	Byeon et al.
9,915,978 B2	3/2018	Dabby et al.		2012/0124532 A1 *	5/2012	Coteus H01L 25/0657
9,934,832 B2	4/2018	Shibata et al.				716/104
9,953,941 B2 *	4/2018	Enquist	H01L 21/50	2012/0136913 A1	5/2012	Duong et al.
10,121,743 B2	11/2018	Kamal		2012/0170345 A1	7/2012	Choi et al.
10,180,692 B2	1/2019	Kouchi		2012/0171818 A1 *	7/2012	Barth H01L 24/80
10,241,150 B2	3/2019	Woo et al.				438/107
10,255,969 B2	4/2019	Eom et al.		2012/0201068 A1	8/2012	Ware
10,262,911 B1	4/2019	Gong		2012/0242346 A1	9/2012	Wang
10,269,586 B2	4/2019	Chou et al.		2012/0256653 A1	10/2012	Cordero et al.
10,289,604 B2	5/2019	Sankaralingam et al.		2012/0262196 A1	10/2012	Yokou
10,354,975 B2 *	7/2019	Soares	H01L 23/585	2012/0286431 A1	11/2012	Foster, Sr. et al.
10,373,657 B2	8/2019	Kondo et al.		2012/0313263 A1	12/2012	Barth
10,446,207 B2	10/2019	Kim et al.		2013/0021866 A1	1/2013	Lee
10,446,601 B2	10/2019	Otake et al.		2013/0032950 A1	2/2013	Ware et al.
10,522,352 B2	12/2019	Delacruz et al.		2013/0051116 A1	2/2013	En et al.
10,580,735 B2	3/2020	Mohammed et al.		2013/0070436 A1 *	3/2013	Sutanto B23K 1/008
10,580,757 B2	3/2020	Nequist et al.				228/248.1
10,580,817 B2	3/2020	Otake et al.		2013/0144542 A1	6/2013	Ernst
10,586,786 B2	3/2020	DeLaCruz et al.		2013/0187292 A1	7/2013	Semmelmeier
10,593,667 B2	3/2020	DeLaCruz et al.		2013/0207268 A1	8/2013	Chapelon
10,600,691 B2	3/2020	DeLaCruz et al.		2013/0242500 A1	9/2013	Lin et al.
10,600,735 B2	3/2020	DeLaCruz et al.		2013/0275823 A1	10/2013	Cordero et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2013/0283008 A1 * 10/2013 Buyuktosunoglu
H01L 25/0657
712/E9.002

2013/0321074 A1 12/2013 Ko et al.

2014/0011324 A1 * 1/2014 Liu H01L 24/74
438/455

2014/0022002 A1 1/2014 Chua-Eoan et al.

2014/0176187 A1 * 6/2014 Jayasena G11C 7/1006
326/39

2014/0177626 A1 * 6/2014 Thottethodi H04L 45/00
257/774

2014/0269022 A1 * 9/2014 Xie H10B 10/12
438/106

2014/0285253 A1 9/2014 Jeon et al.

2014/0323046 A1 10/2014 Asai

2014/0369148 A1 12/2014 Matsui et al.

2014/0379975 A1 * 12/2014 Ikegami G06F 1/3296
711/104

2015/0032962 A1 * 1/2015 Buyuktosunoglu
G06F 12/0811
711/122

2015/0121052 A1 * 4/2015 Emma G06F 9/44
713/1

2015/0162220 A1 * 6/2015 Chou H01L 24/81
438/124

2015/0199997 A1 * 7/2015 Ito G11C 7/20
327/77

2015/0228584 A1 8/2015 Huang et al.

2015/0262902 A1 9/2015 Shen et al.

2015/0348943 A1 * 12/2015 Chen H01L 21/76877
257/734

2015/0370754 A1 * 12/2015 Morimoto G06F 1/20
361/679.31

2016/0111386 A1 * 4/2016 England H01L 24/03
438/107

2016/0155725 A1 * 6/2016 Chen H01L 25/50
257/770

2016/0217087 A1 * 7/2016 Lim G06F 13/4282

2016/0225431 A1 8/2016 Best et al.

2016/0233134 A1 8/2016 Lim et al.

2016/0276313 A1 * 9/2016 Kume H01L 23/481

2016/0329312 A1 11/2016 O'Mullan et al.

2016/0379115 A1 12/2016 Burger

2017/0092615 A1 3/2017 Oyamada

2017/0092616 A1 3/2017 Su et al.

2017/0141079 A1 * 5/2017 Kao H01L 24/80

2017/0148737 A1 5/2017 Fasano et al.

2017/0194309 A1 7/2017 Evans et al.

2017/0213787 A1 7/2017 Alfano et al.

2017/0227605 A1 8/2017 Kim et al.

2017/0278213 A1 9/2017 Eckert

2017/0278789 A1 9/2017 Chuang

2017/0285584 A1 10/2017 Nakagawa et al.

2017/0301625 A1 10/2017 Mahajan et al.

2018/0017614 A1 1/2018 Leedy

2018/0068218 A1 3/2018 Yoo et al.

2018/0286800 A1 10/2018 Kamal et al.

2018/0330992 A1 11/2018 Delacruz et al.

2018/0330993 A1 11/2018 Delacruz et al.

2018/0331037 A1 11/2018 Mohammed et al.

2018/0331038 A1 11/2018 Delacruz et al.

2018/0331072 A1 11/2018 Nequist et al.

2018/0331094 A1 11/2018 Delacruz et al.

2018/0331095 A1 11/2018 Delacruz et al.

2018/0350775 A1 12/2018 DeLaCruz

2018/0373975 A1 12/2018 Yu et al.

2018/0374788 A1 12/2018 Nakagawa et al.

2019/0006322 A1 1/2019 Park

2019/0042377 A1 2/2019 Teig et al.

2019/0042912 A1 2/2019 Teig et al.

2019/0042929 A1 2/2019 Teig et al.

2019/0043832 A1 2/2019 Teig et al.

2019/0051641 A1 2/2019 Lee et al.

2019/0109057 A1 4/2019 Hargan et al.

2019/0123022 A1 4/2019 Teig et al.

2019/0123023 A1 4/2019 Teig et al.

2019/0123024 A1 4/2019 Teig et al.

2019/0156215 A1 5/2019 Matveev et al.

2019/0180183 A1 6/2019 Diamant et al.

2019/0214991 A1 7/2019 Ngai

2019/0244933 A1 8/2019 Or-Bach

2020/0013699 A1 1/2020 Liu

2020/0143866 A1 5/2020 Biswas et al.

2020/0194052 A1 6/2020 Shaeffer et al.

2020/0203318 A1 6/2020 Nequist et al.

2020/0219771 A1 7/2020 DeLaCruz et al.

2020/0227389 A1 7/2020 Teig et al.

2020/0273798 A1 8/2020 Mohammed et al.

2020/0293872 A1 9/2020 Teig et al.

2020/0294858 A1 9/2020 DeLaCruz et al.

2020/0372345 A1 11/2020 Li

FOREIGN PATENT DOCUMENTS

EP 3698402 A1 8/2020

JP 2010250511 A * 11/2010 G06F 12/08

KR 20150137970 A 12/2015

WO 2017138121 A1 8/2017

WO 2019079625 A1 4/2019

WO 2019079631 A1 4/2019

OTHER PUBLICATIONS

Author Unknown, "Vector Supercomputer SX Series: SX-Aurora TSUBASA," Oct. 2017, 2 pages, NEC Corporation.

Bansal, Samta, "3D-IC is Now Real: Wide-IO is Driving 3D-IC TSV," Cadence Flash Memory Summit, Aug. 2012, 14 pages, Cadence Design Systems, Inc.

Black, Bryan, "Die Stacking is Happening!," Dec. 9, 2013, 53 pages, Advanced Micro Devices, Inc., Santa Clara, California.

Black, Bryan, et al., "3D Processing Technology and its Impact on iA32 Microprocessors," Proceedings of 2004 IEEE International Conference on Computer Design: VLSI in Computers and Processors, Oct. 11-13, 2004, 3 pages, IEEE, San Jose, California.

Black, Bryan, et al., "Die Stacking (3D) Microarchitecture," Proceedings of the 39th Annual IEEE/ACM International Symposium on Microarchitecture, Dec. 9-13, 2006, 11 pages, IEEE, Orlando, Florida.

Hajkazemi, Mohammad Hossein, et al., "Wide I/O or LPDDR? Exploration and Analysis of Performance, Power and Temperature Trade-offs of Emerging DRAM Technologies in Embedded MPSoCs," Proceedings of 33rd IEEE International Conference on Computer Design (ICCD), Oct. 18-21, 2015, 8 pages, IEEE, New York City, New York.

Invitation to Pay Additional Fees for Commonly Owned International Patent Application PCT/US2018/056559, dated Feb. 7, 2019, 15 pages, International Searching Authority (EPO).

Invitation to Pay Additional Fees for Commonly Owned International Patent Application PCT/US2018/056565, dated Feb. 12, 2019, 13 pages, International Searching Authority (EPO).

Loh, Gabriel H., et al., "Processor Design in 3D Die-Stacking Technologies," IEEE Micro, May/Jun. 2007, 18 pages, vol. 27, Issue 3, IEEE Computer Society.

Nakamoto, Mark, et al., "Simulation Methodology and Flow Integration for 3D IC Stress Management," 2010 IEEE Custom Integrated Circuits Conference, Sep. 19-22, 2010, 4 pages, IEEE, San Jose, CA, USA.

Tran, Kevin, et al., "Start Your HBM/2.5D Design Today," High-Bandwidth Memory White Paper, Mar. 29, 2016, 6 pages, Amkor Technology, Inc., San Jose, CA, USA.

Wu, Xiaoxia, et al., "Electrical Characterization for Intertier Connections and Timing Analysis for 3-D ICs," IEEE Transactions on Very Large Scale Integration (VLSI) Systems, Dec. 6, 2010, 5 pages, IEEE.

Kim, Jung-Sik, et al., "A 1.2 V 12.8 GB/s 2 GB Mobile Wide-I/O DRAM With 4x128 I/Os Using TSV Based Stacking," IEEE Journal of Solid-State Circuits, Jan. 2012, 10 pages, vol. 47, No. 1, IEEE.

(56)

References Cited

OTHER PUBLICATIONS

International Search Report and Written Opinion of Commonly Owned International Patent Application PCT/US2018/056565, mailed Apr. 2, 2019, 17 pages, International Searching Authority (European Patent Office).

Non-Published Commonly Owned Related U.S. Appl. No. 16/806,854, filed Mar. 2, 2020, 75 pages, Xcelsis Corporation.

Non-Published Commonly Owned Related U.S. Appl. No. 16/889,698, filed Jun. 1, 2020, 74 pages, Xcelsis Corporation.

Non-Published Commonly Owned Related U.S. Appl. No. 16/891,027, filed Jun. 2, 2020, 47 pages, Xcelsis Corporation.

International Search Report and Written Opinion of Commonly Owned International Patent Application PCT/US2018/056559, dated Mar. 29, 2019, 17 pages, International Searching Authority (European Patent Office).

Non-Published Commonly Owned Related U.S. Appl. No. 16/827,467, filed Mar. 23, 2020, 74 pages, Xcelsis Corporation.

Nair, Vinod and Hinton, Geoffrey E., "Rectified linear units improve restricted Boltzmann machines," ICML, pp. 807-814, 2010.

He, Kaiming, Zhang, Xiangyu, Ren, Shaoqing, and Sun, Jian, "Delving deep into rectifiers: Surpassing human-level performance on imagenet classification," arXiv preprint arXiv: 1502.01852, 2015, pp. 1026-1034.

Non-Published Commonly Owned U.S. Appl. No. 16/806,934, filed Mar. 2, 2020, 74 pages, Xcelsis Corporation.

Yuang, Zhang, "Key Research Issues of Memory Architecture for Three Dimensional Multi-Core Processors," Department of Physics, Nanjing University (May 2015).

A. Jouve et al., "1 μ m Pitch direct hybrid bonding with <300nm wafer-to-wafer overlay accuracy," 2017 IEEE SOI-3D-Subthreshold Microelectronics Technology Unified Conference (S3S), Burlingame, CA, USA, 2017, pp. 1-2.

* cited by examiner

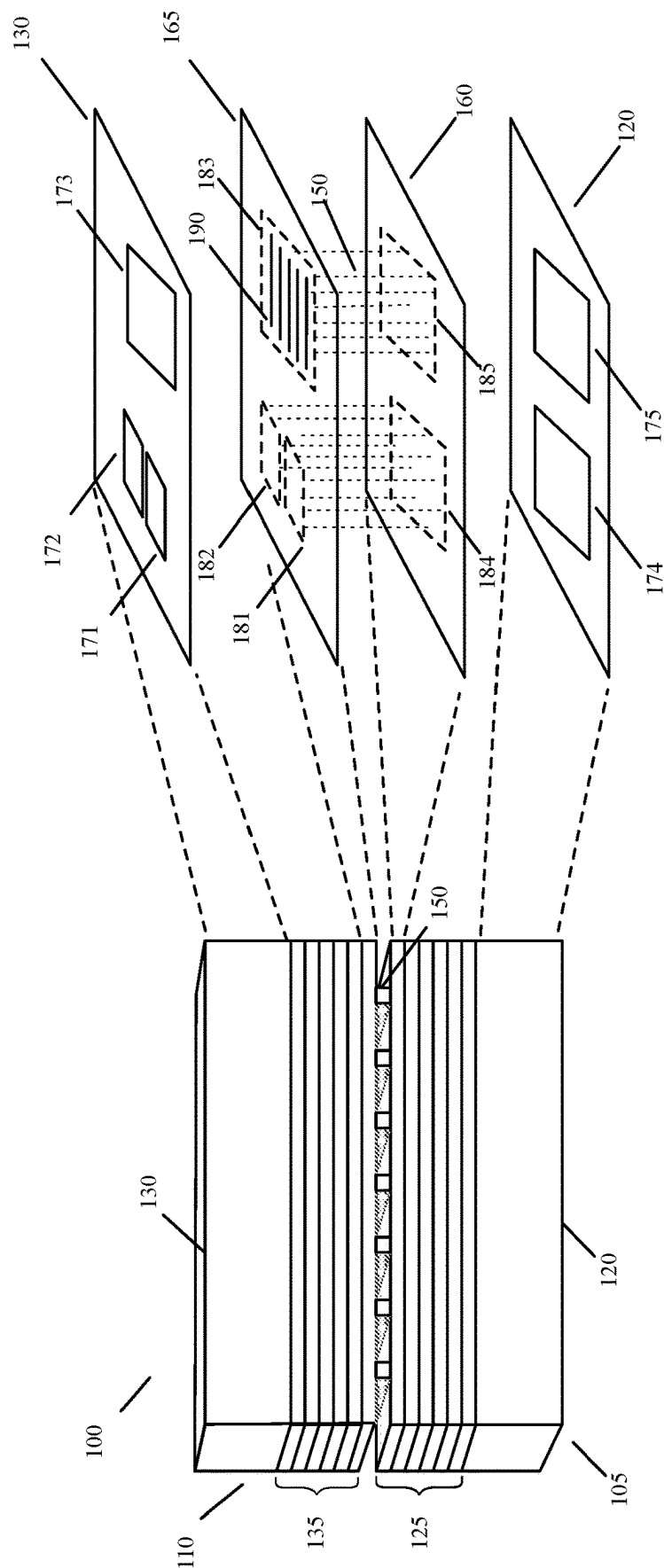


Figure 1

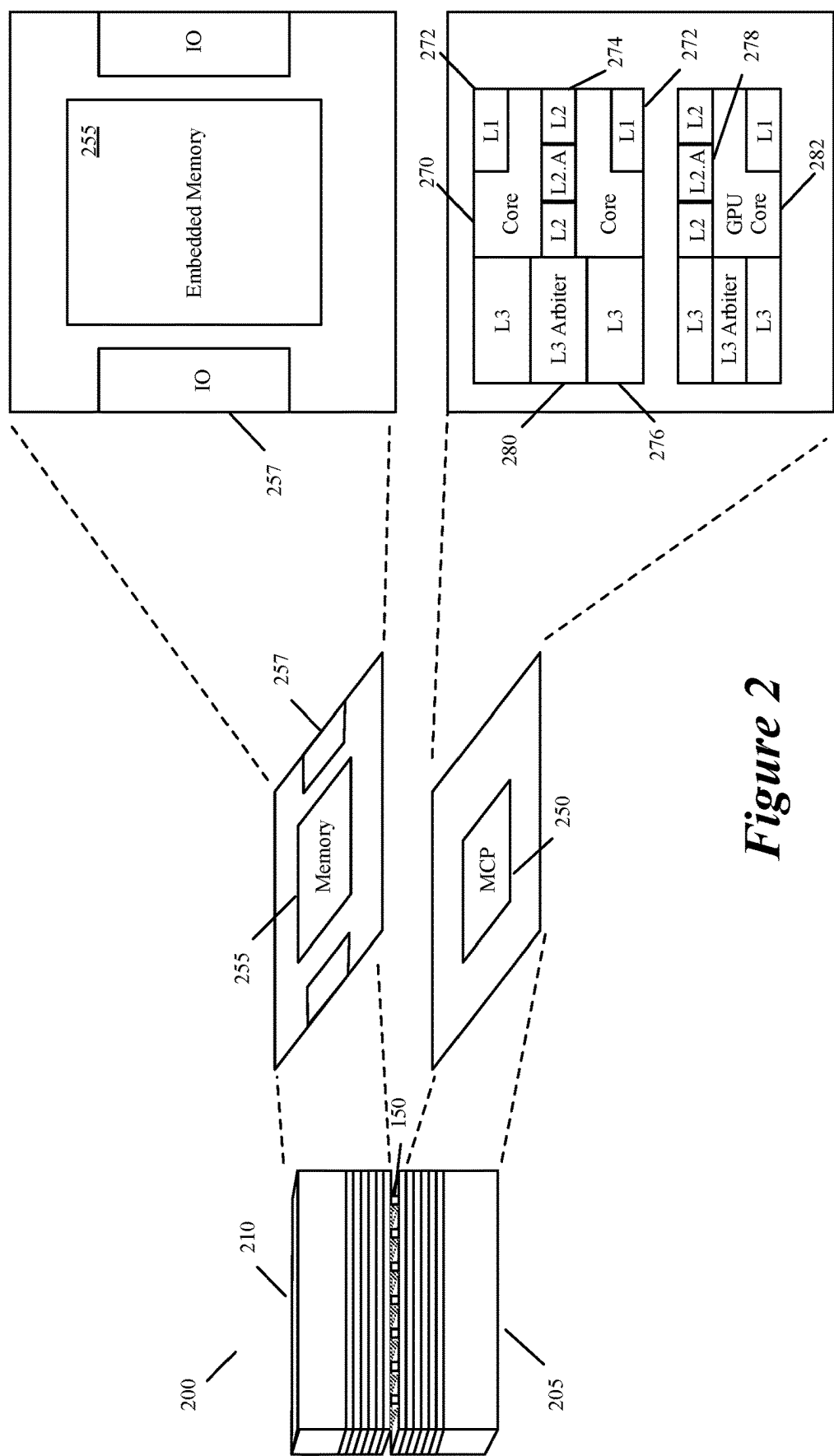


Figure 2

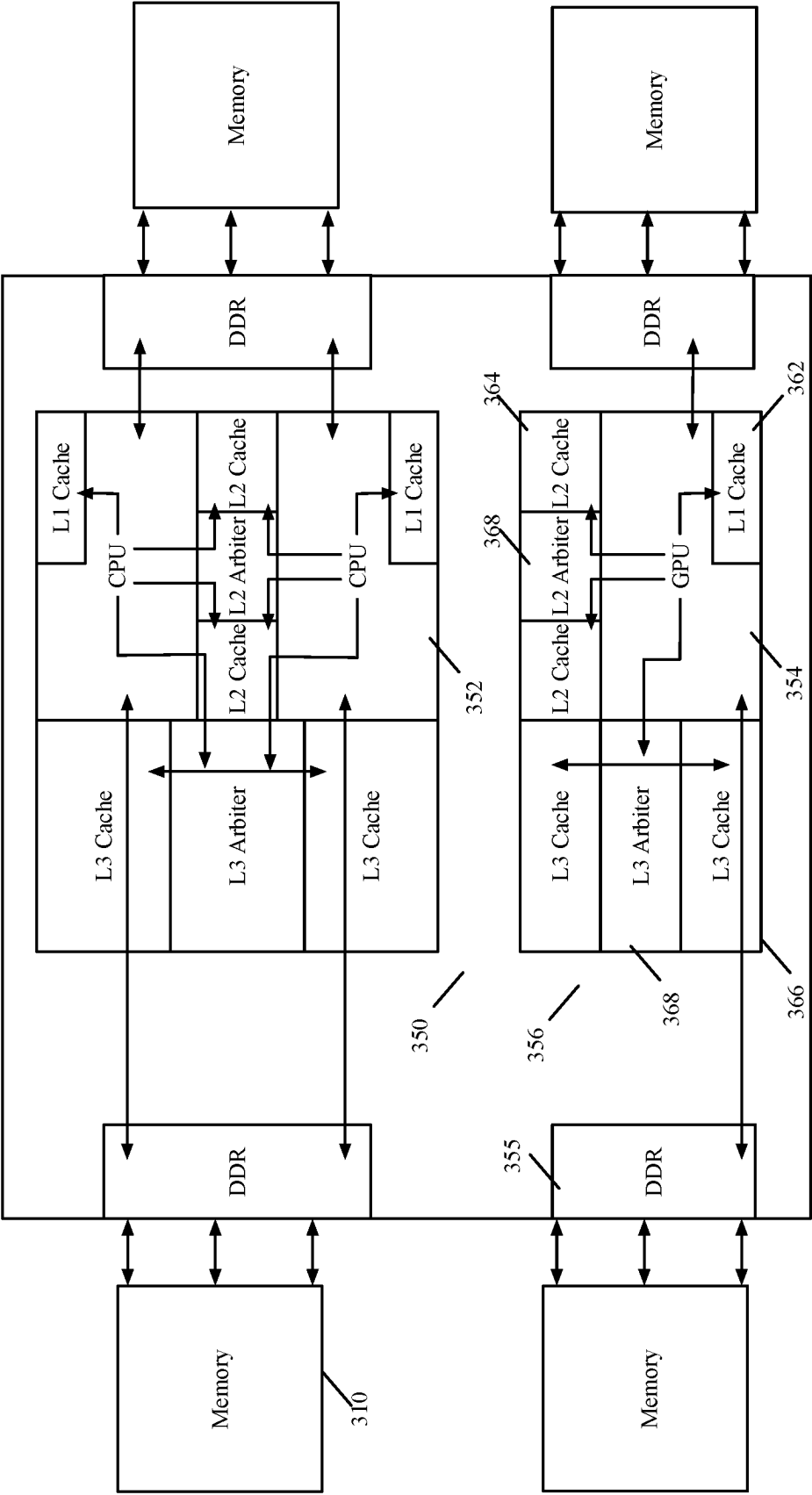


Figure 3
Prior Art

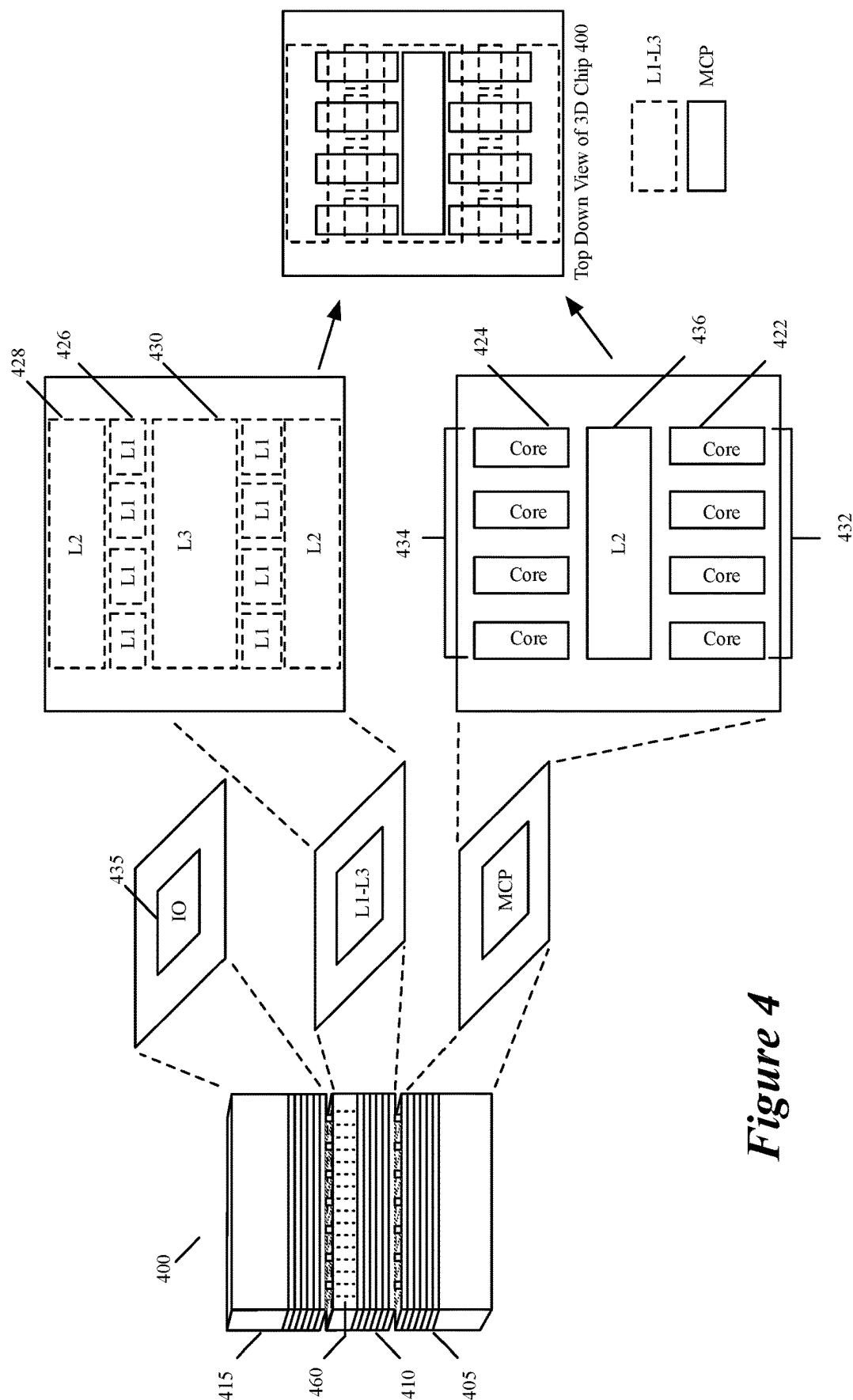


Figure 4

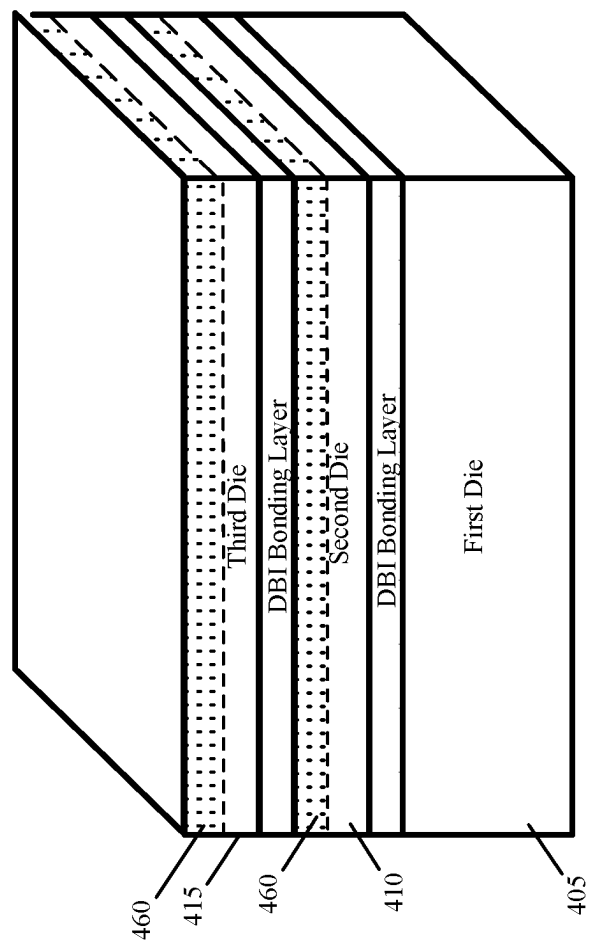


Figure 5

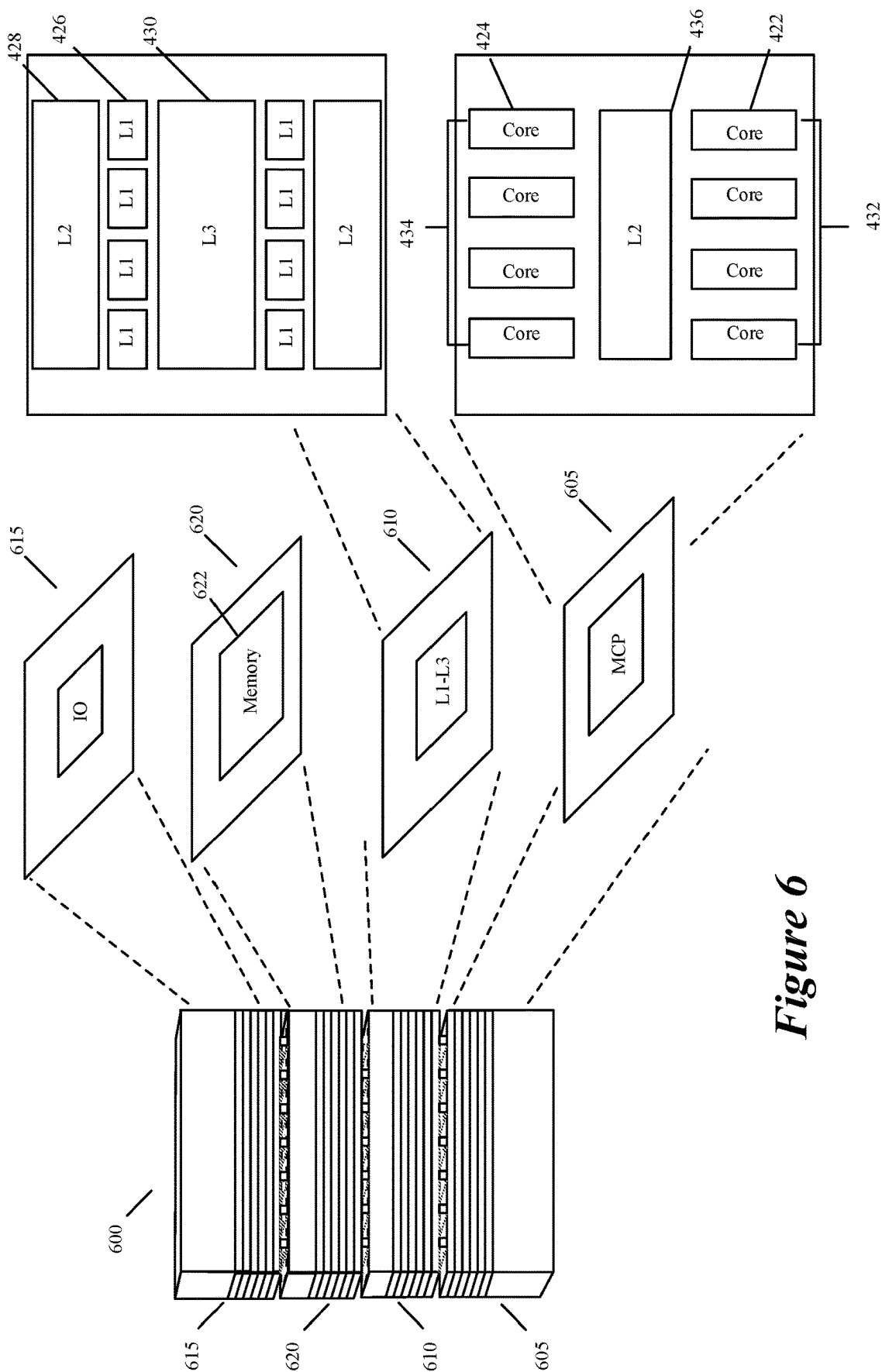


Figure 6

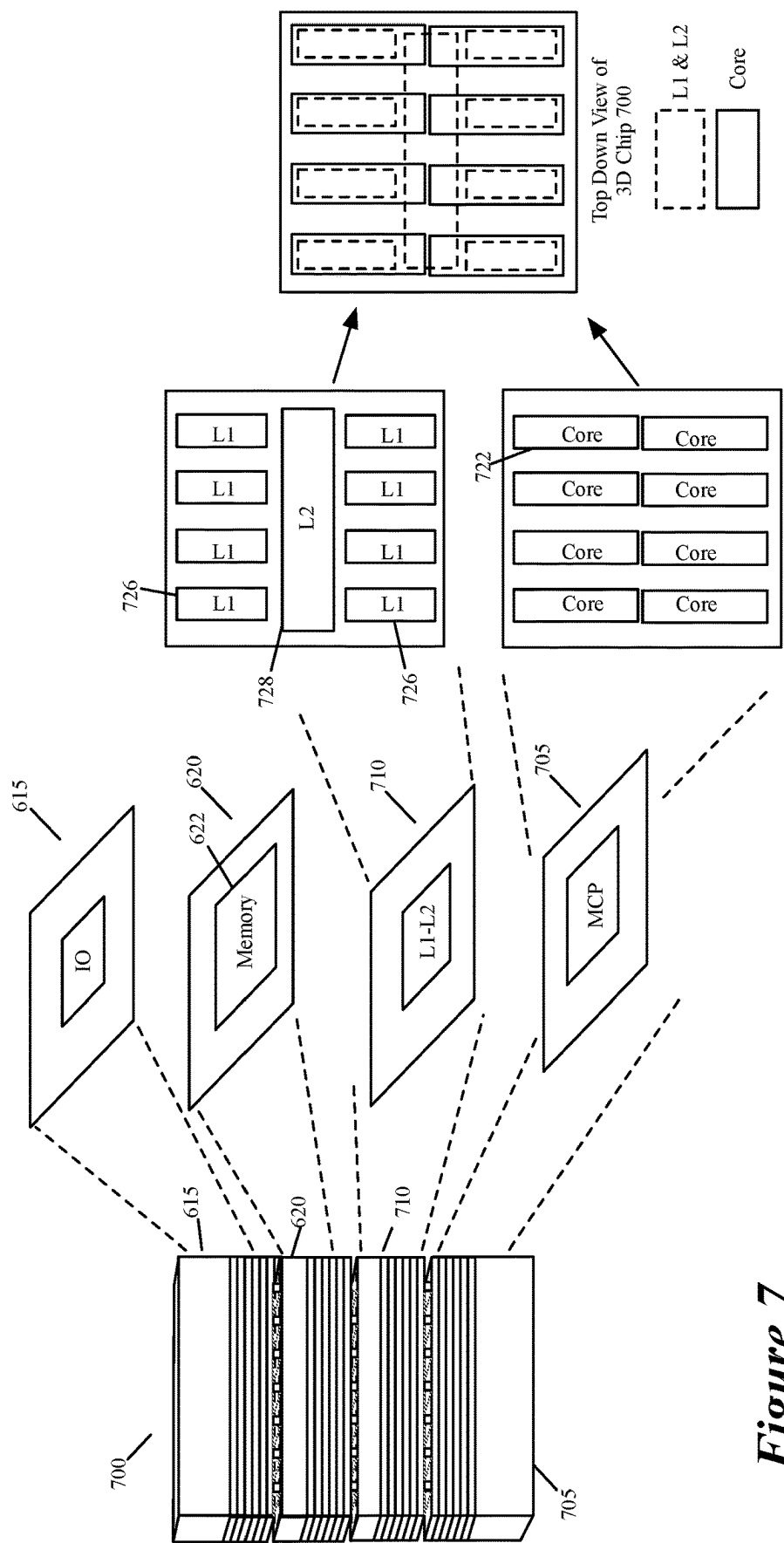


Figure 7

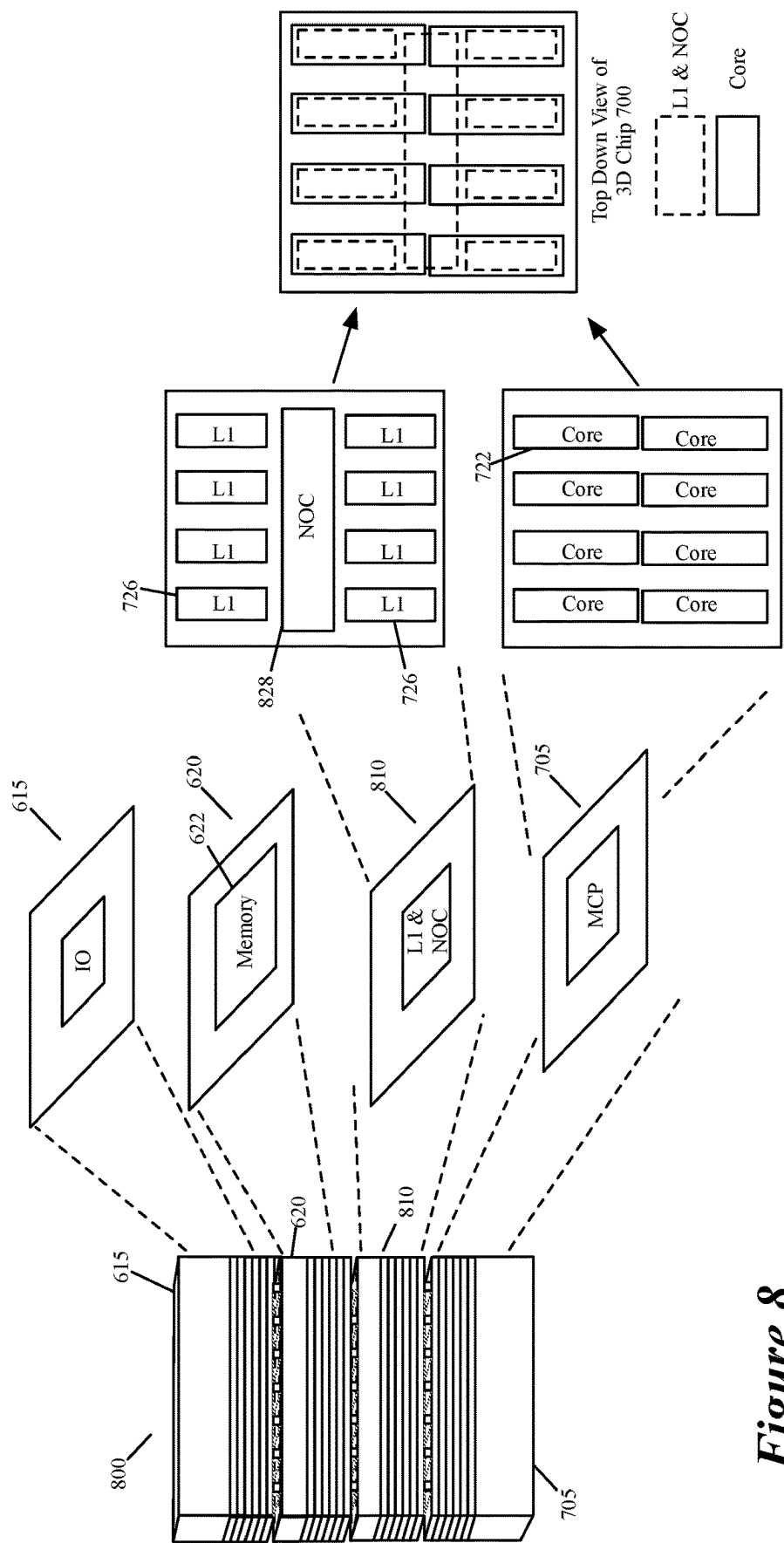


Figure 8

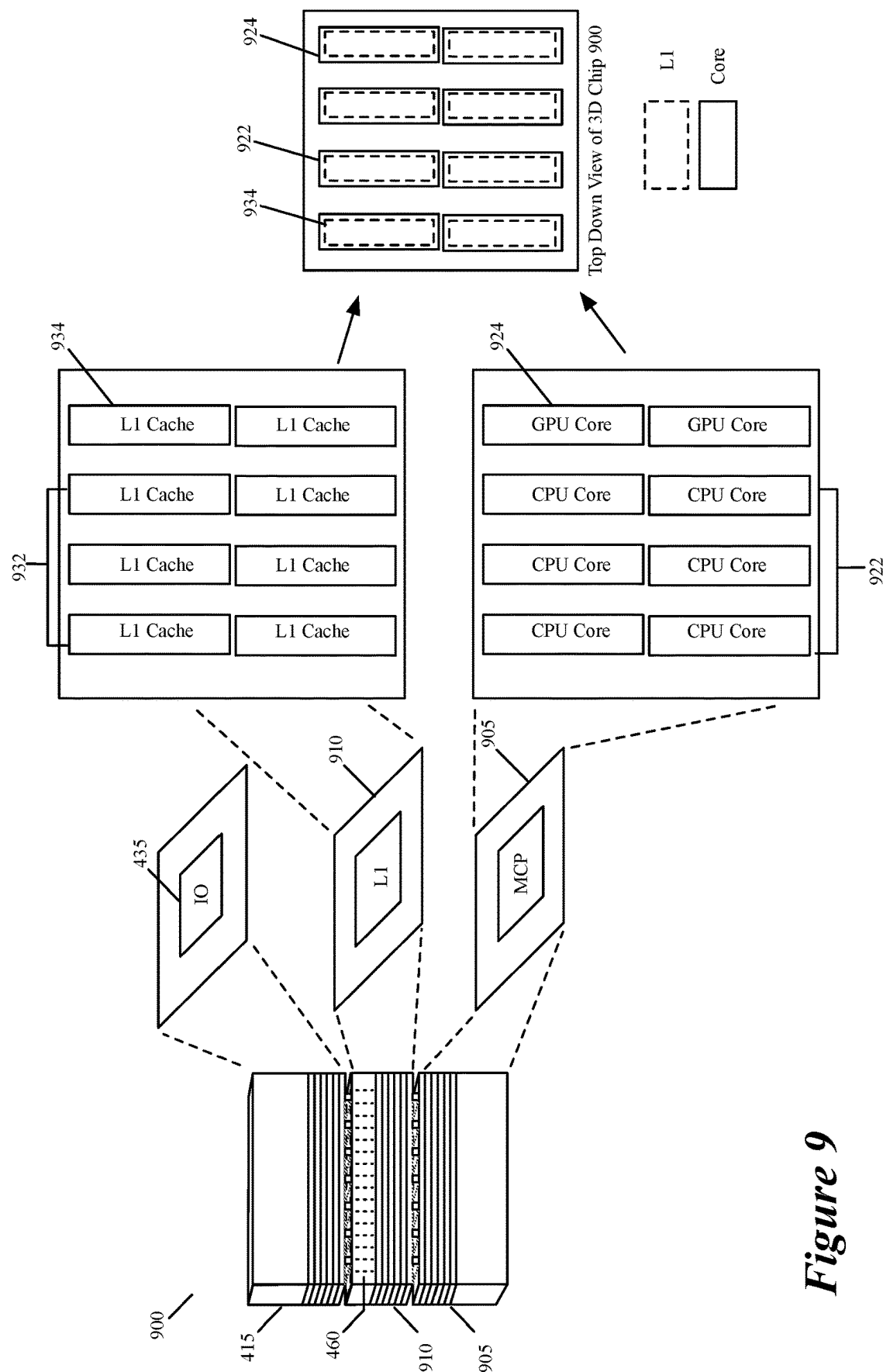


Figure 9

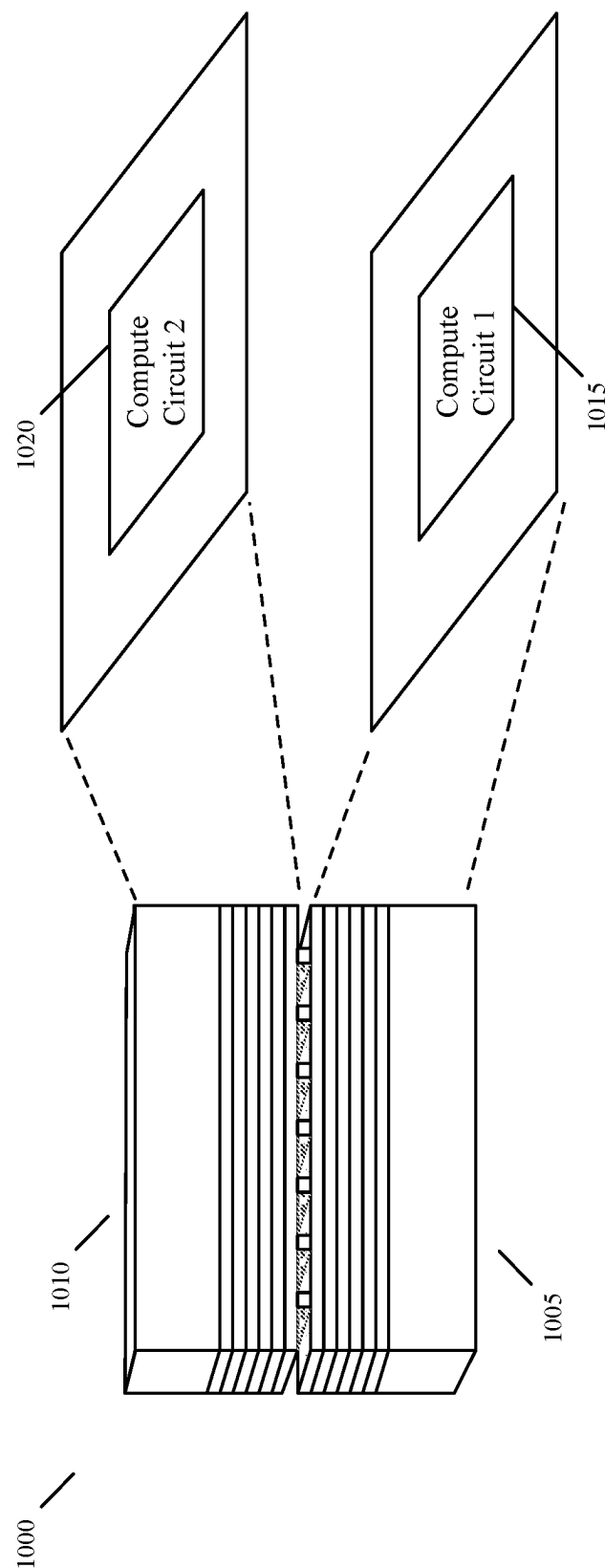


Figure 10

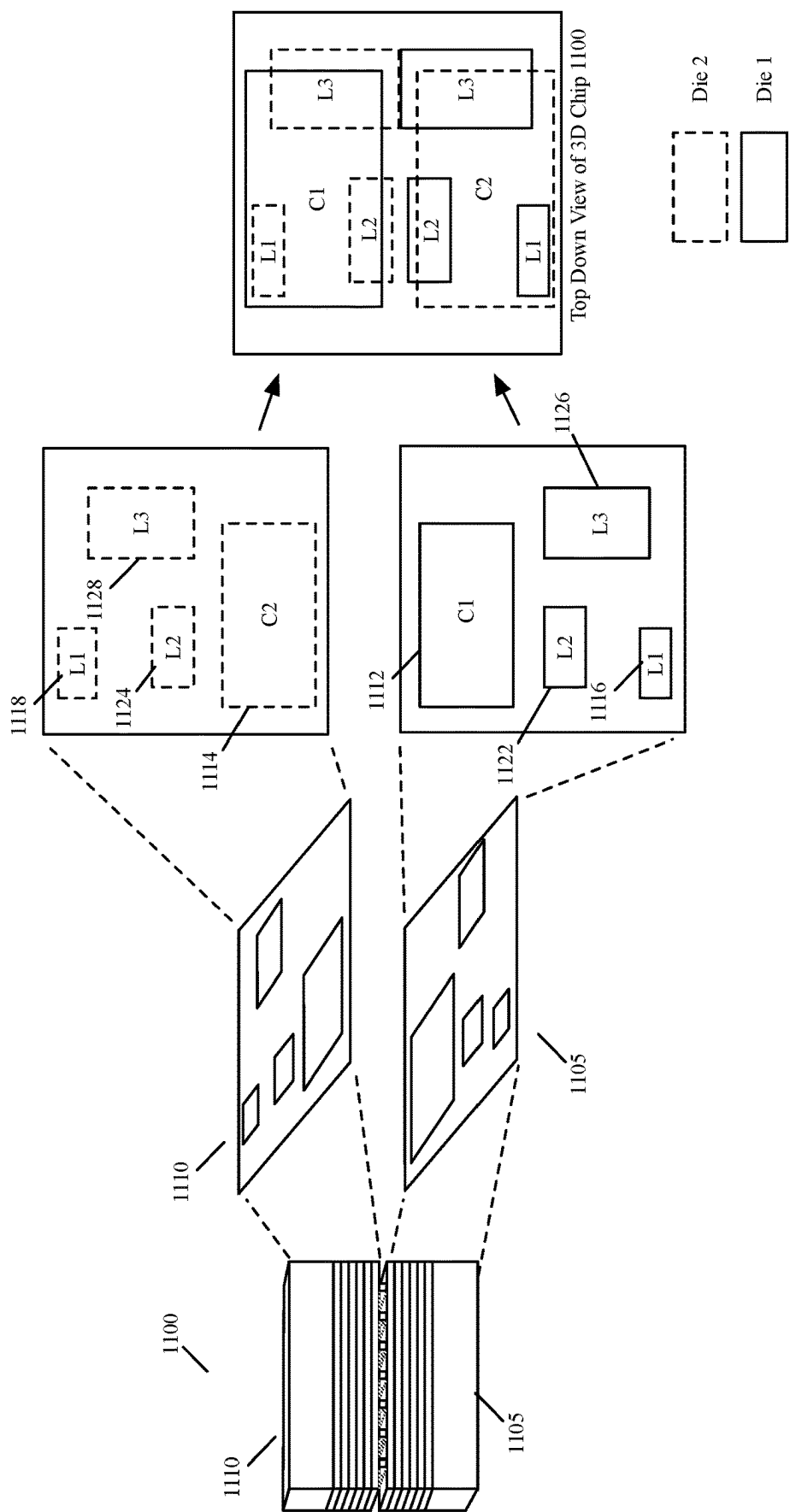


Figure 11

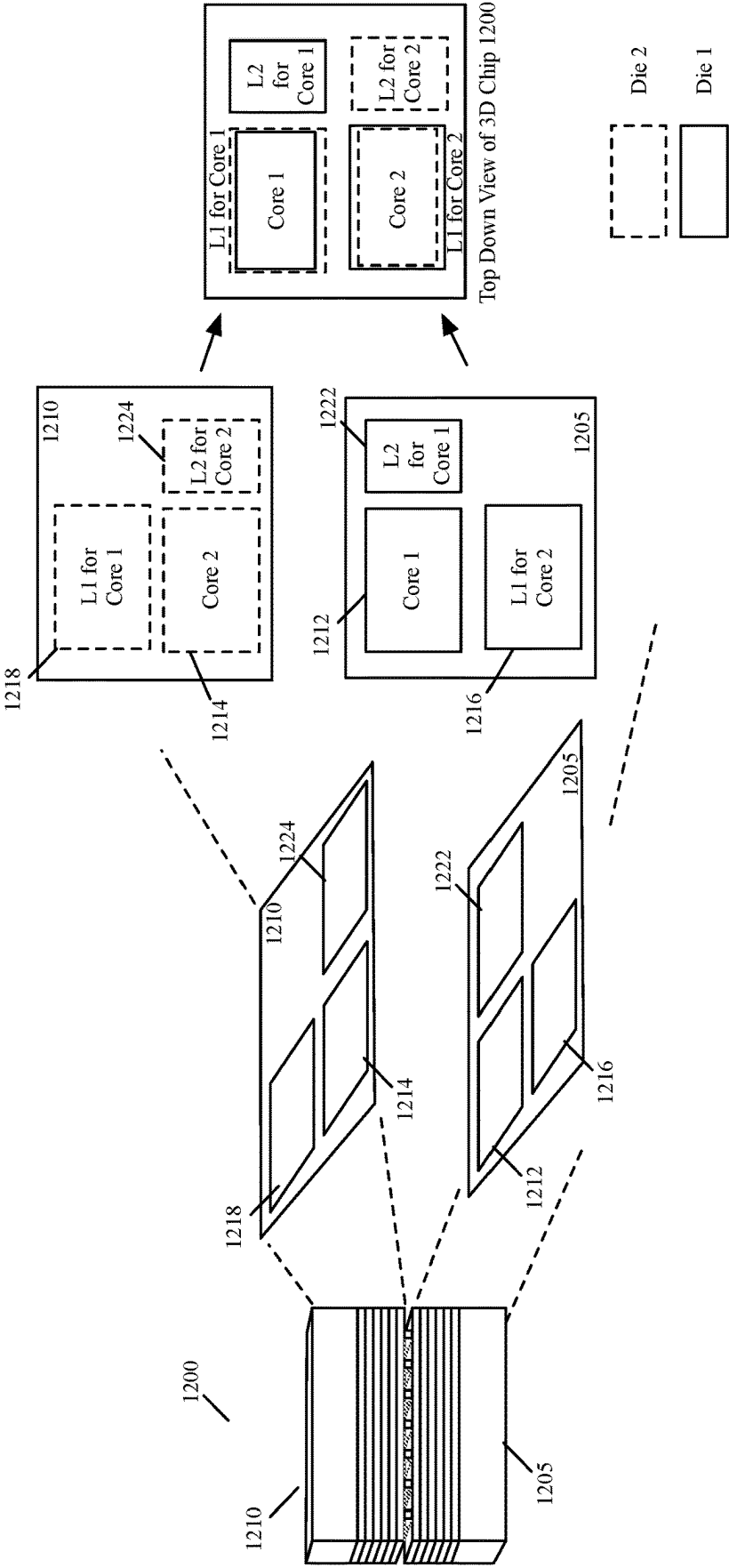


Figure 12

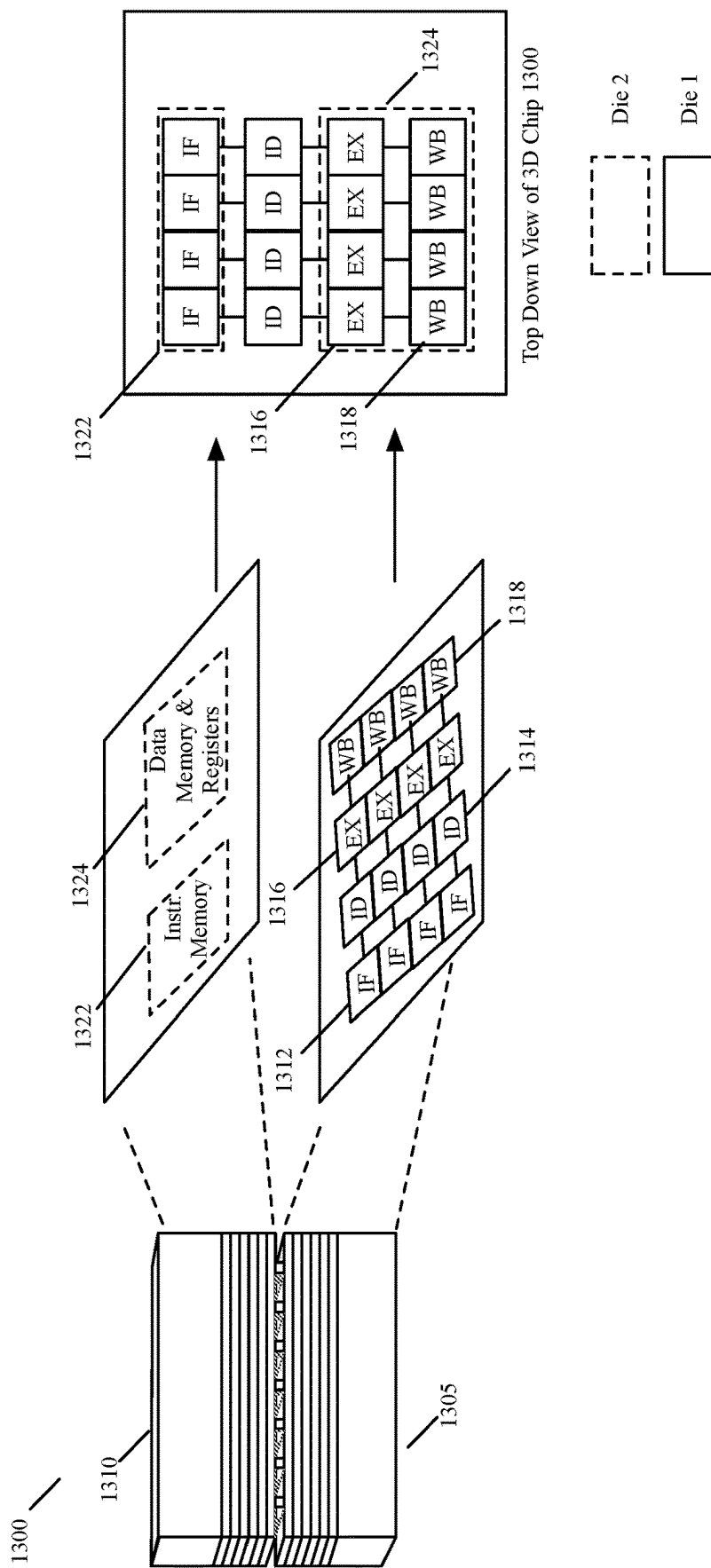


Figure 13

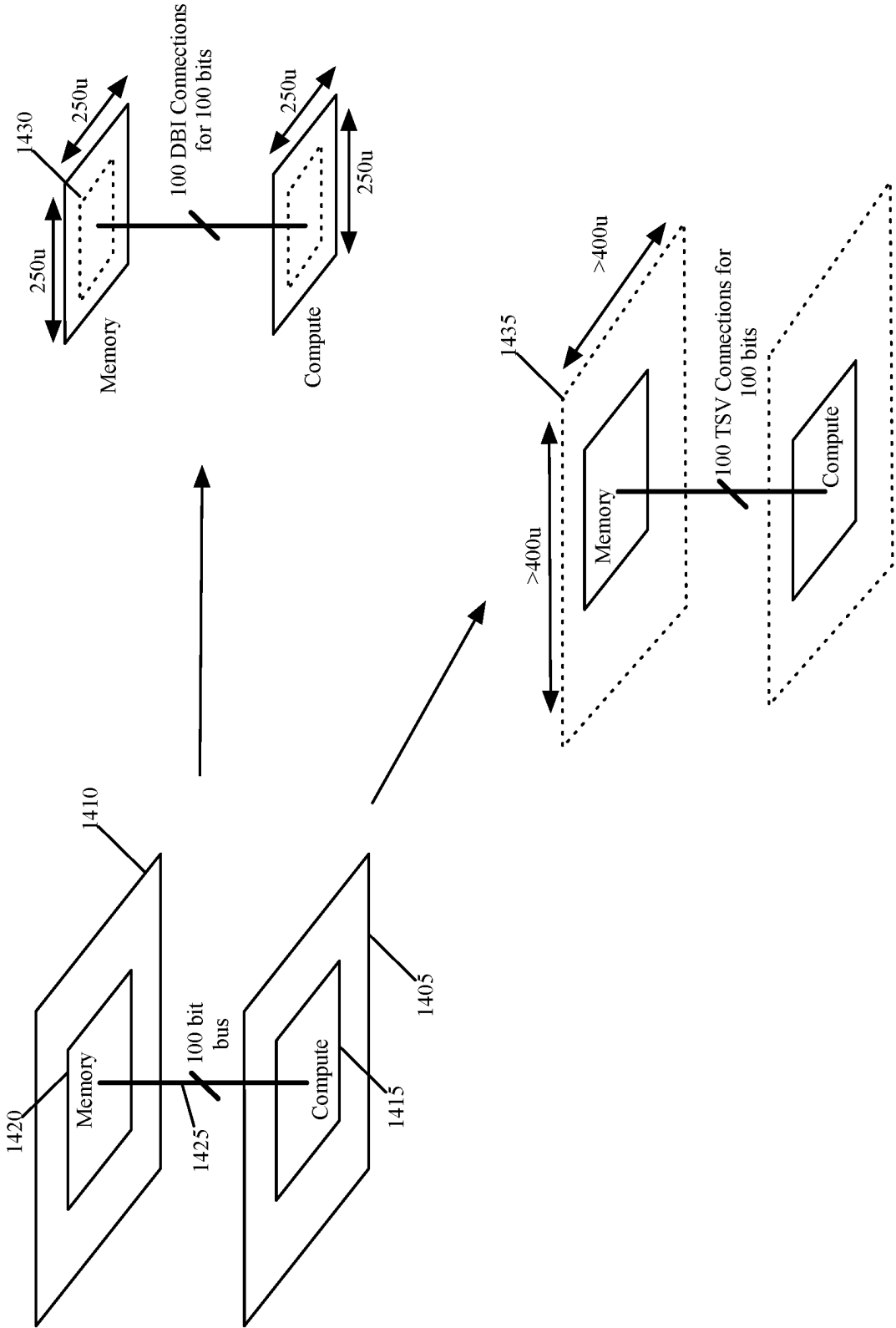


Figure 14

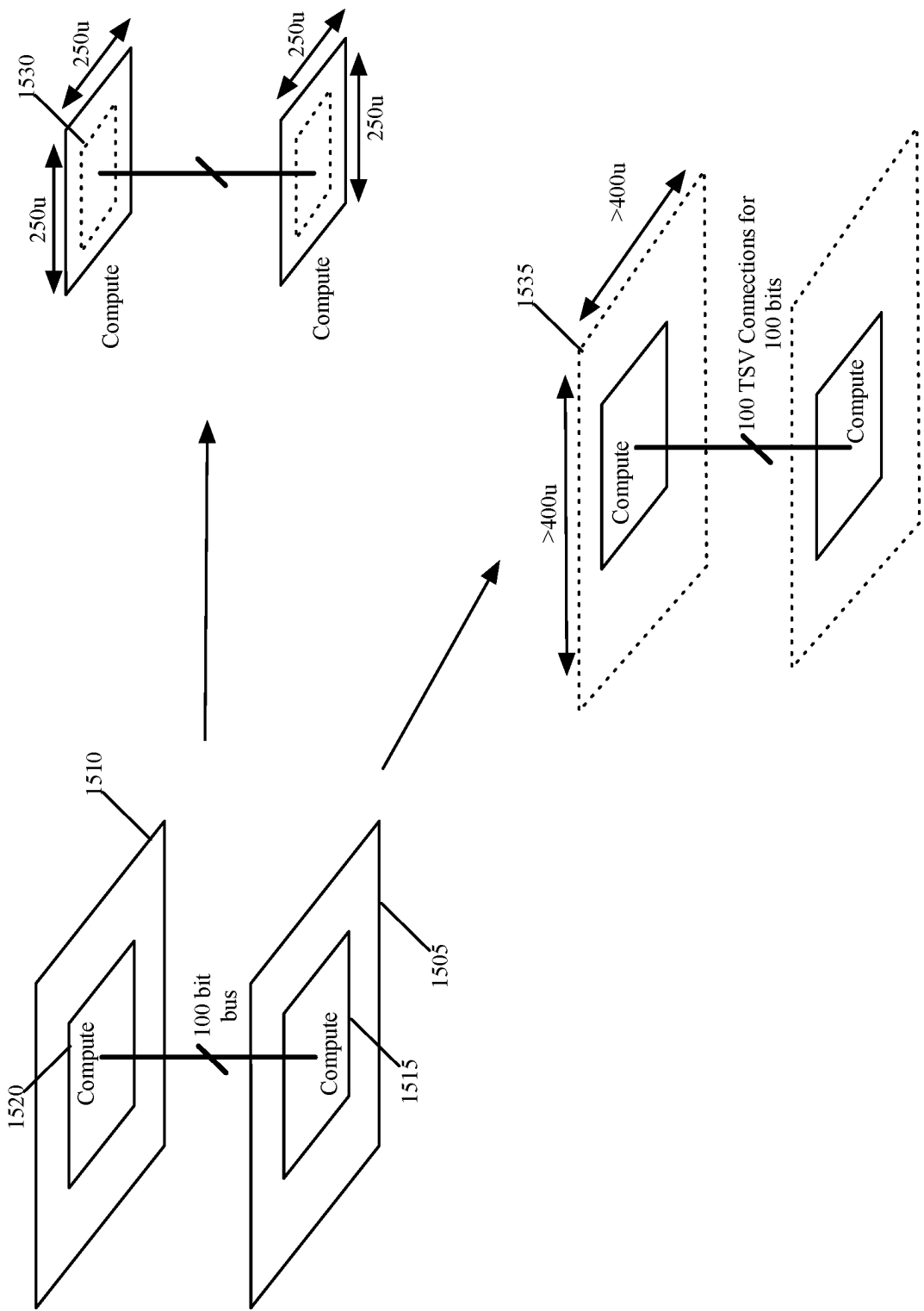


Figure 15

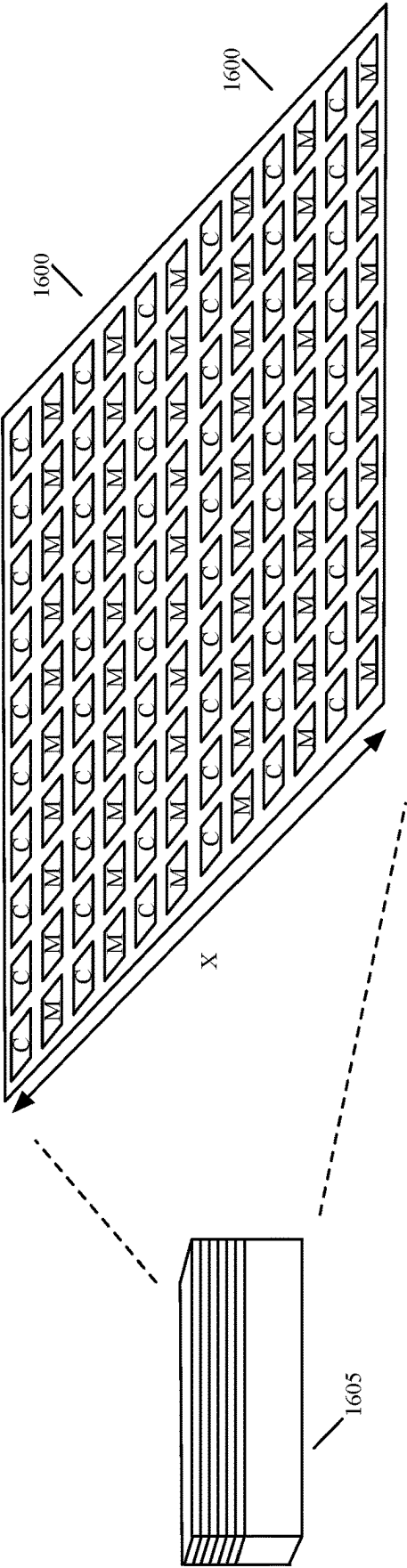


Figure 17
Prior Art

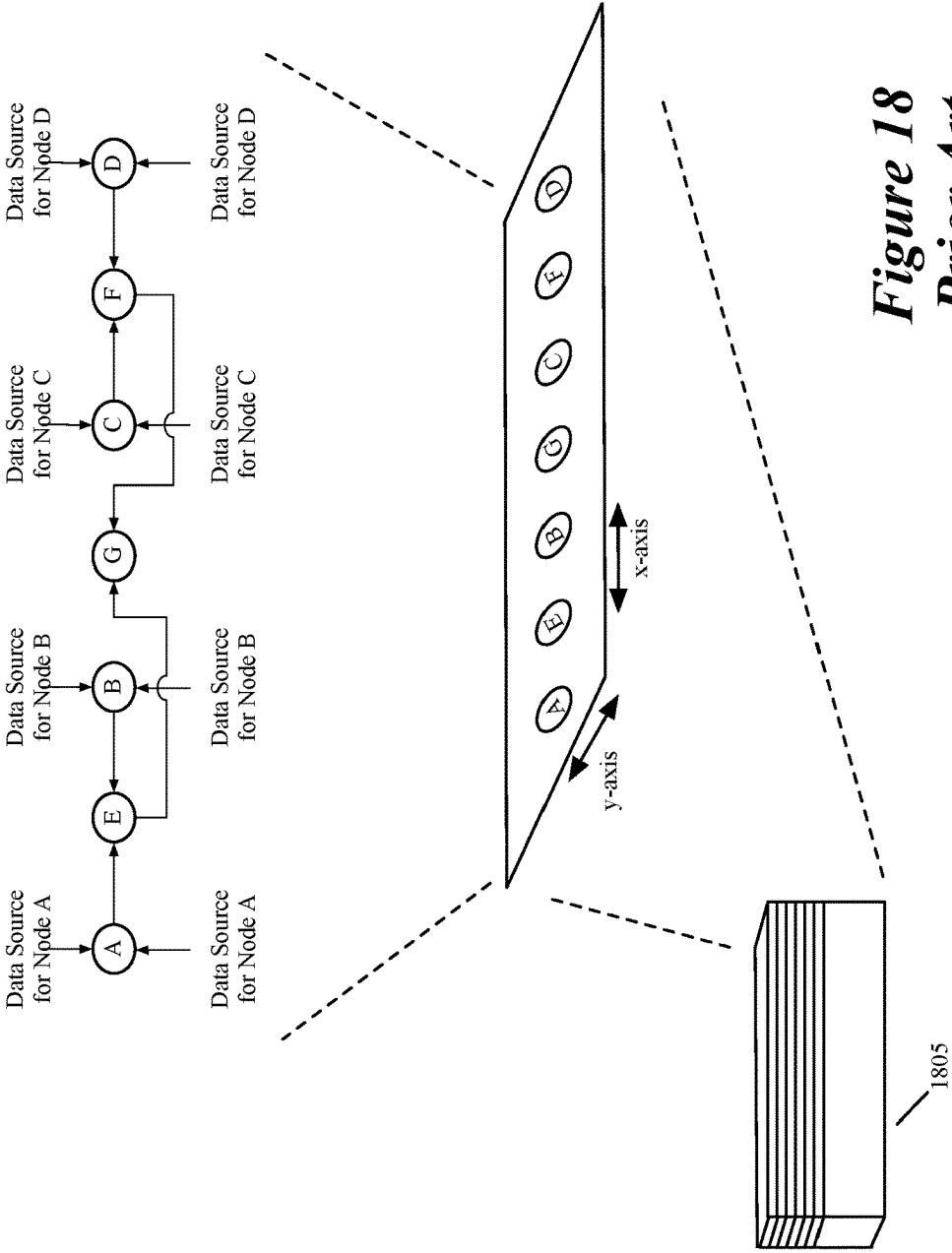
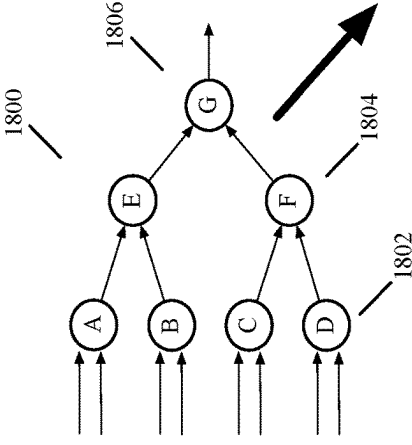


Figure 18
Prior Art



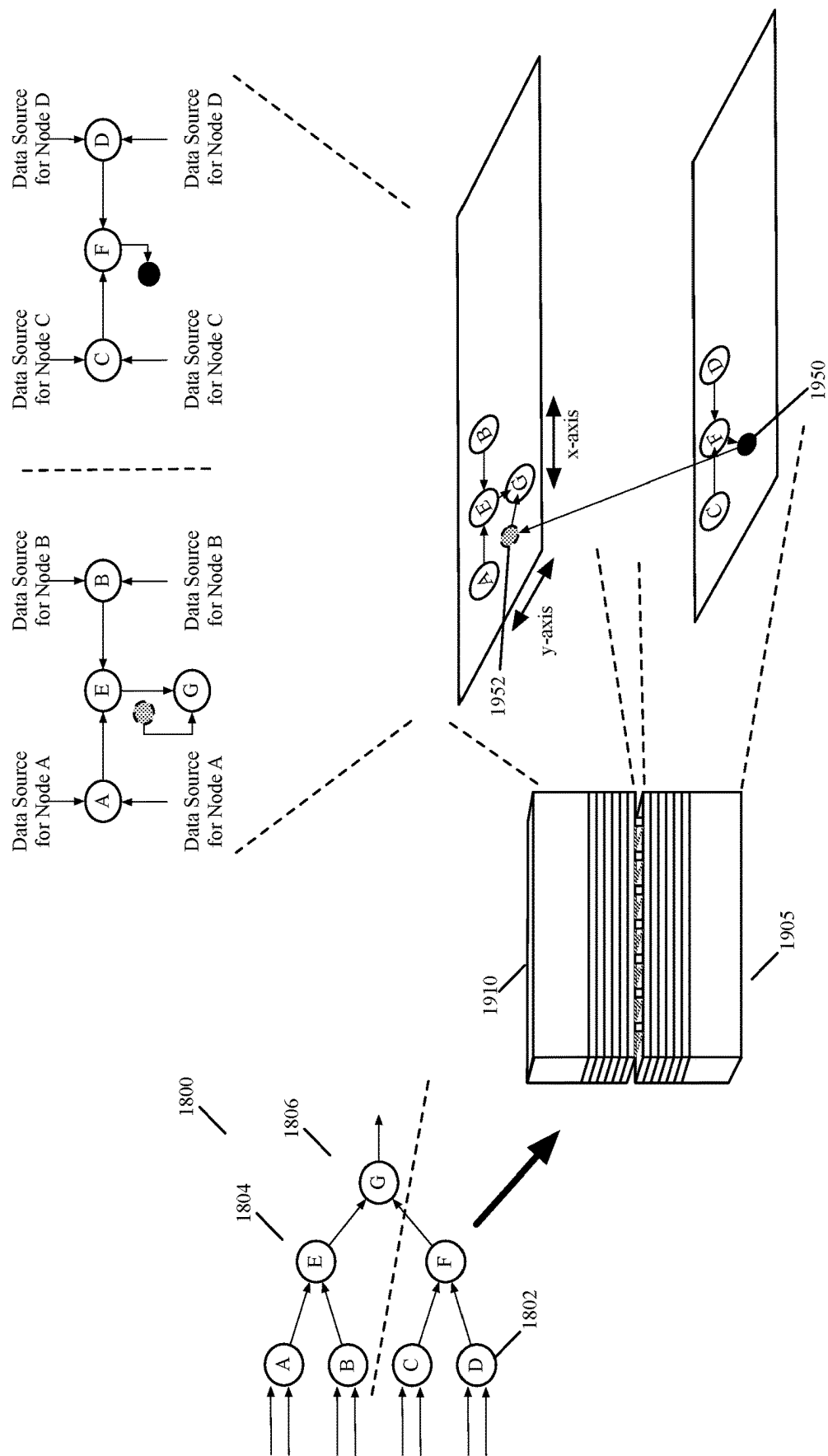


Figure 19

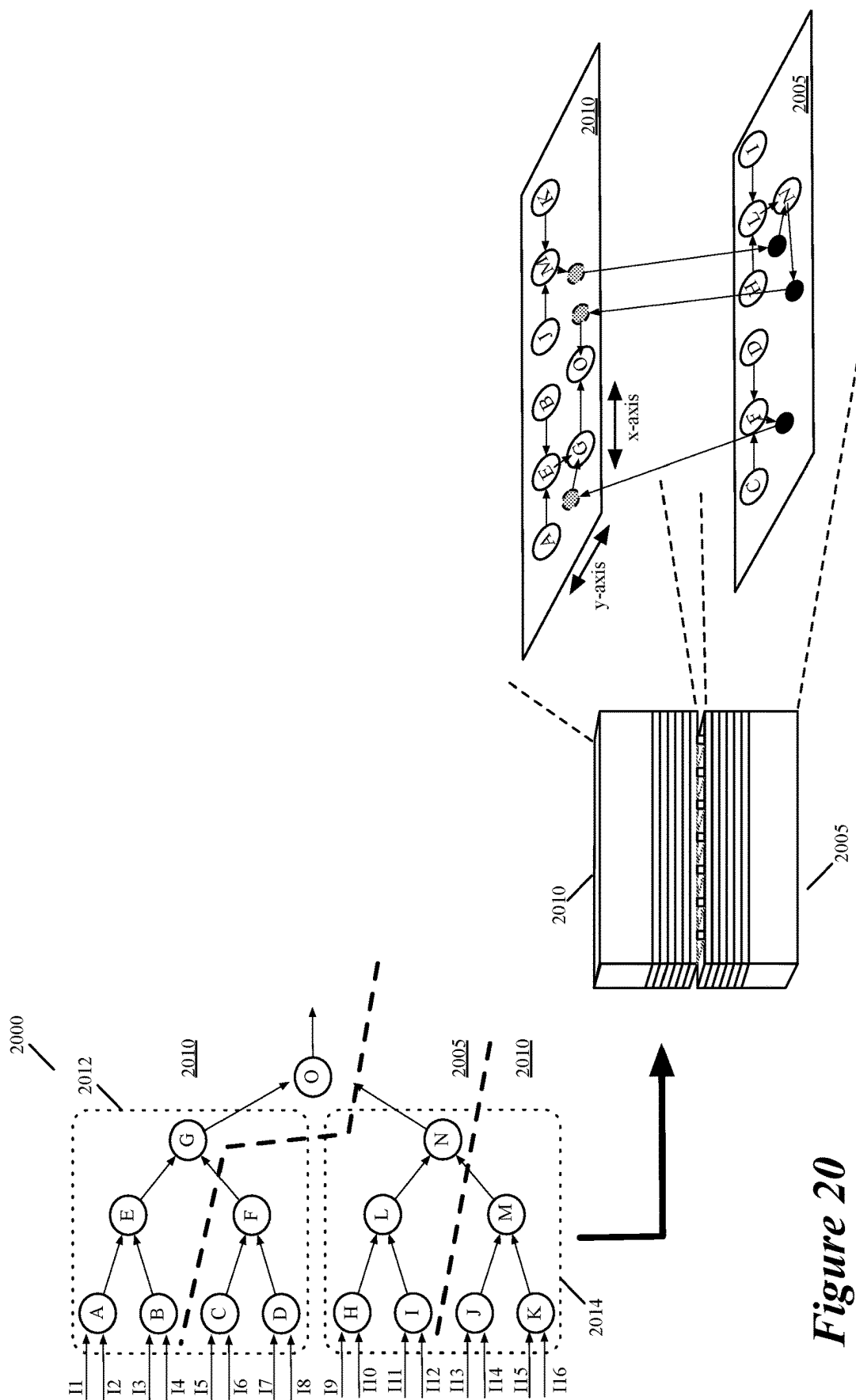


Figure 20

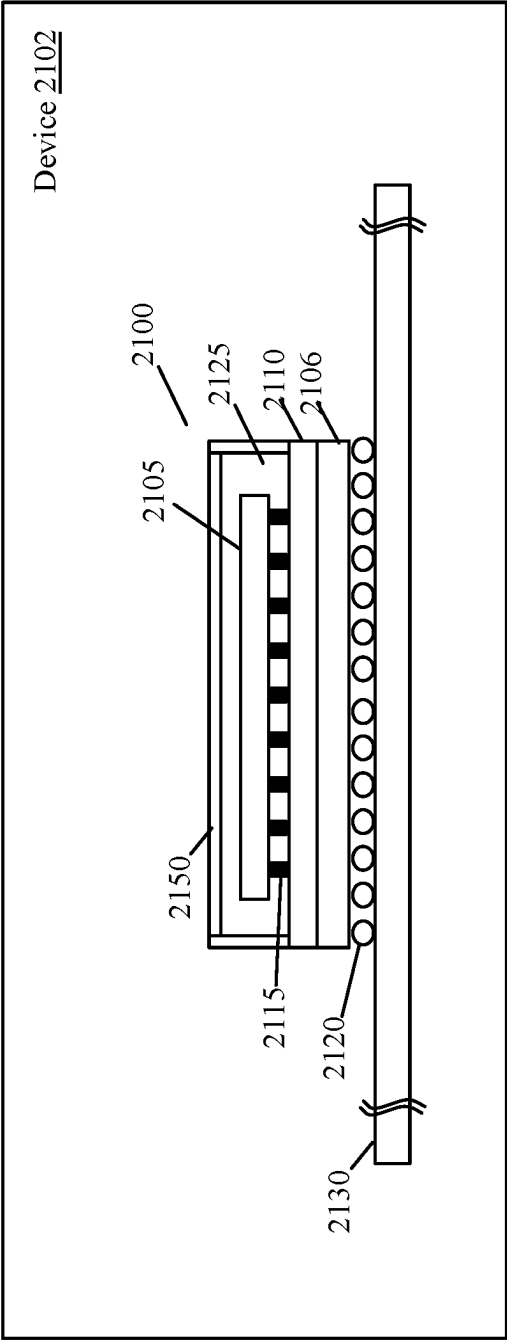


Figure 21

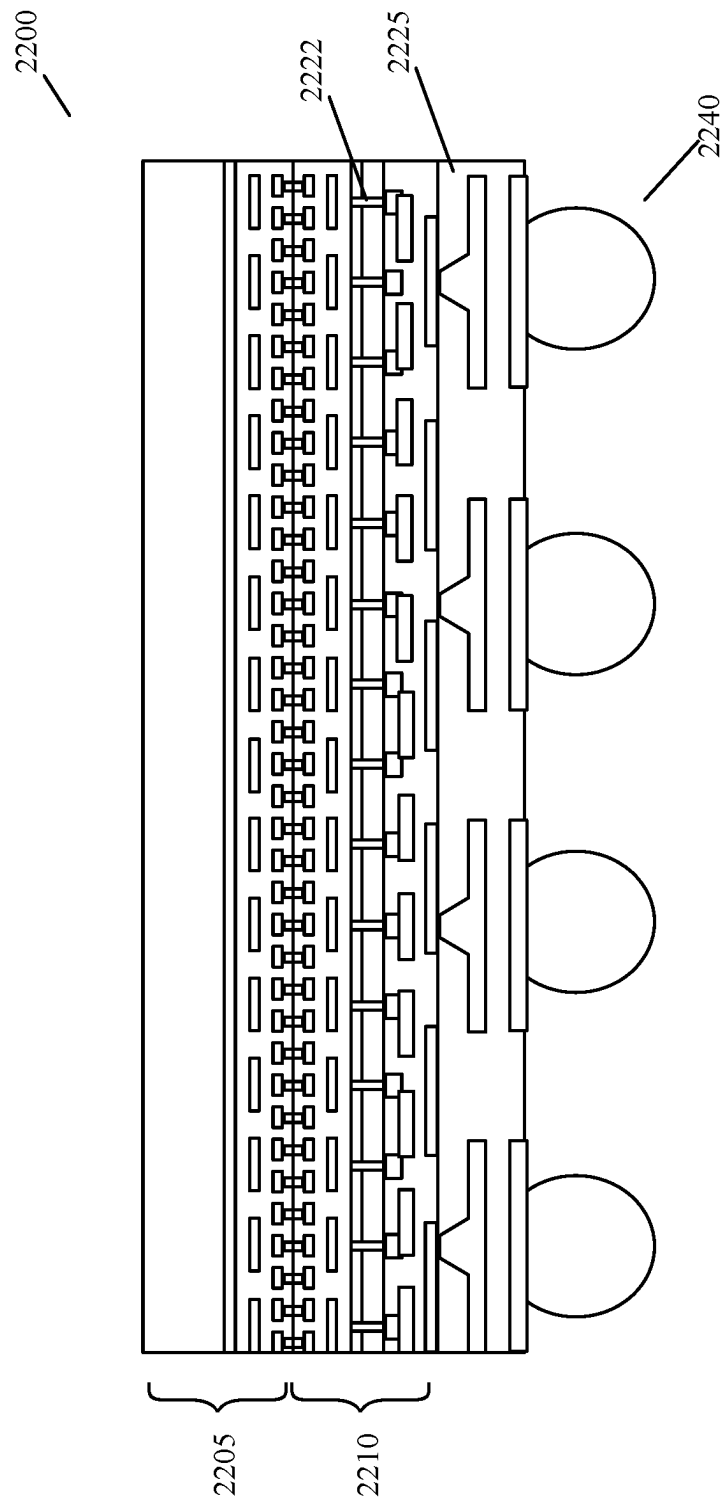
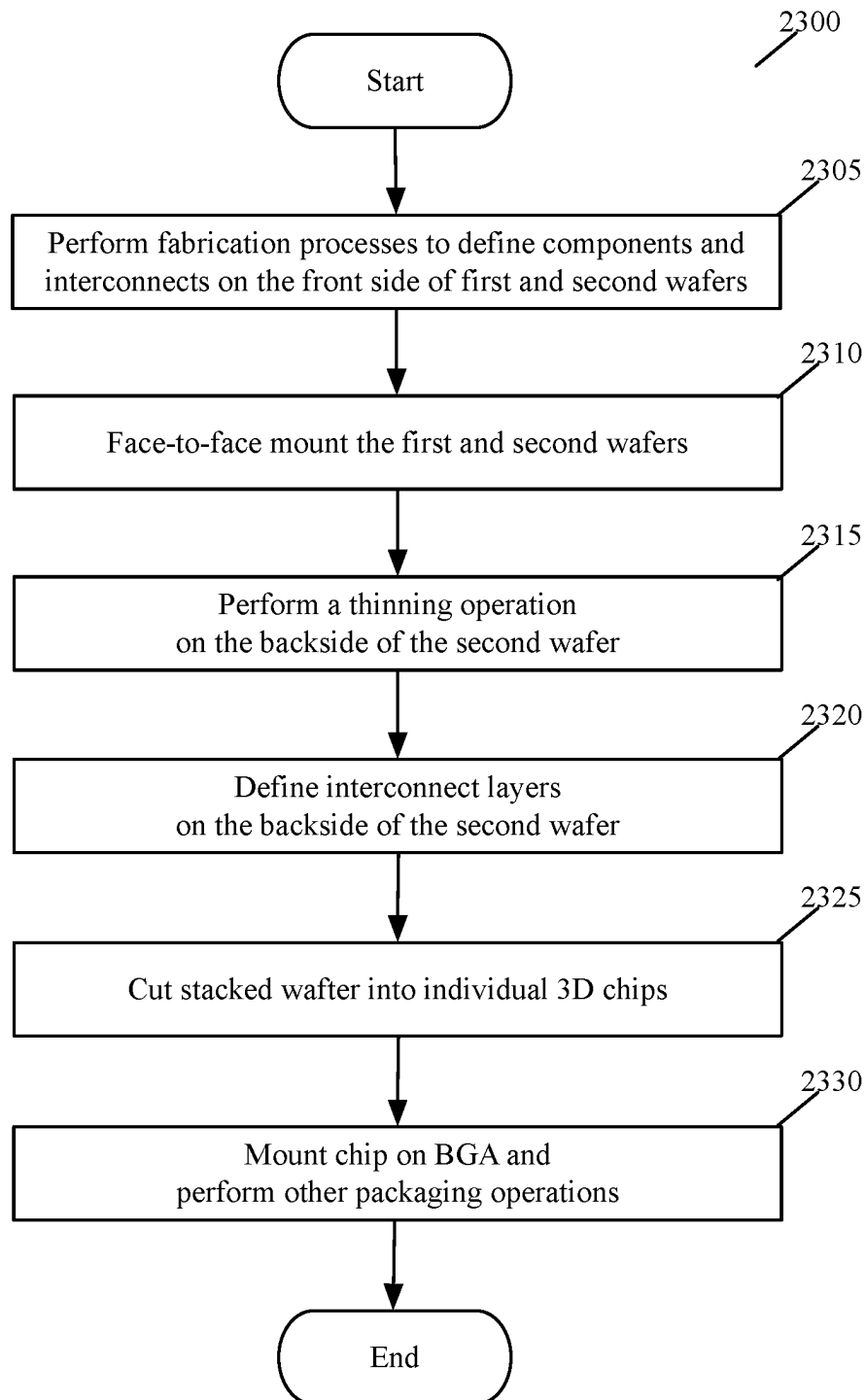


Figure 22

***Figure 23***

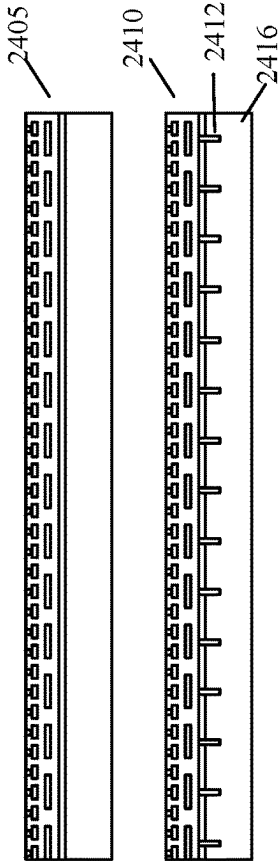


Figure 24

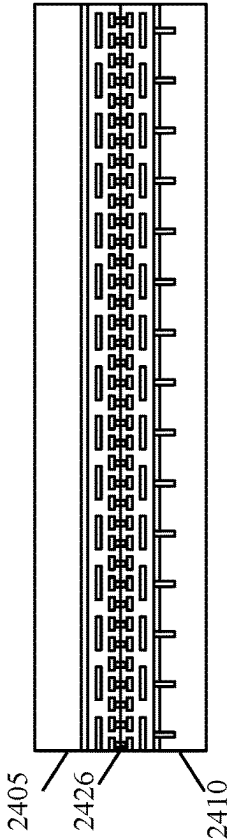


Figure 25

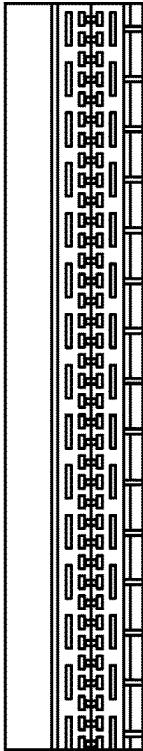


Figure 26

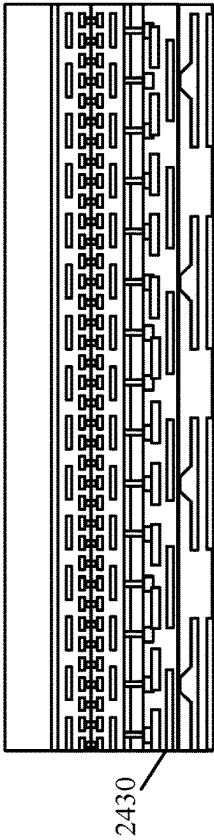


Figure 27

2800

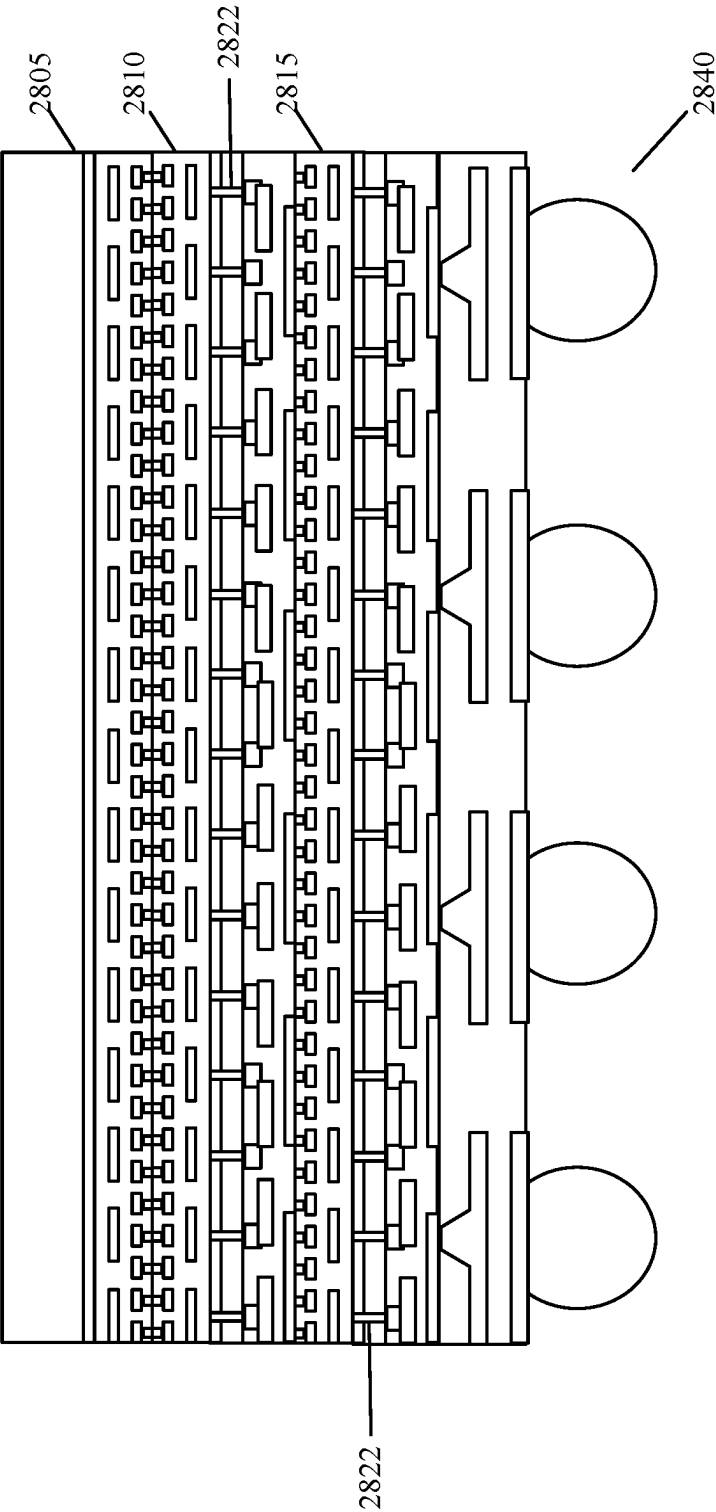


Figure 28

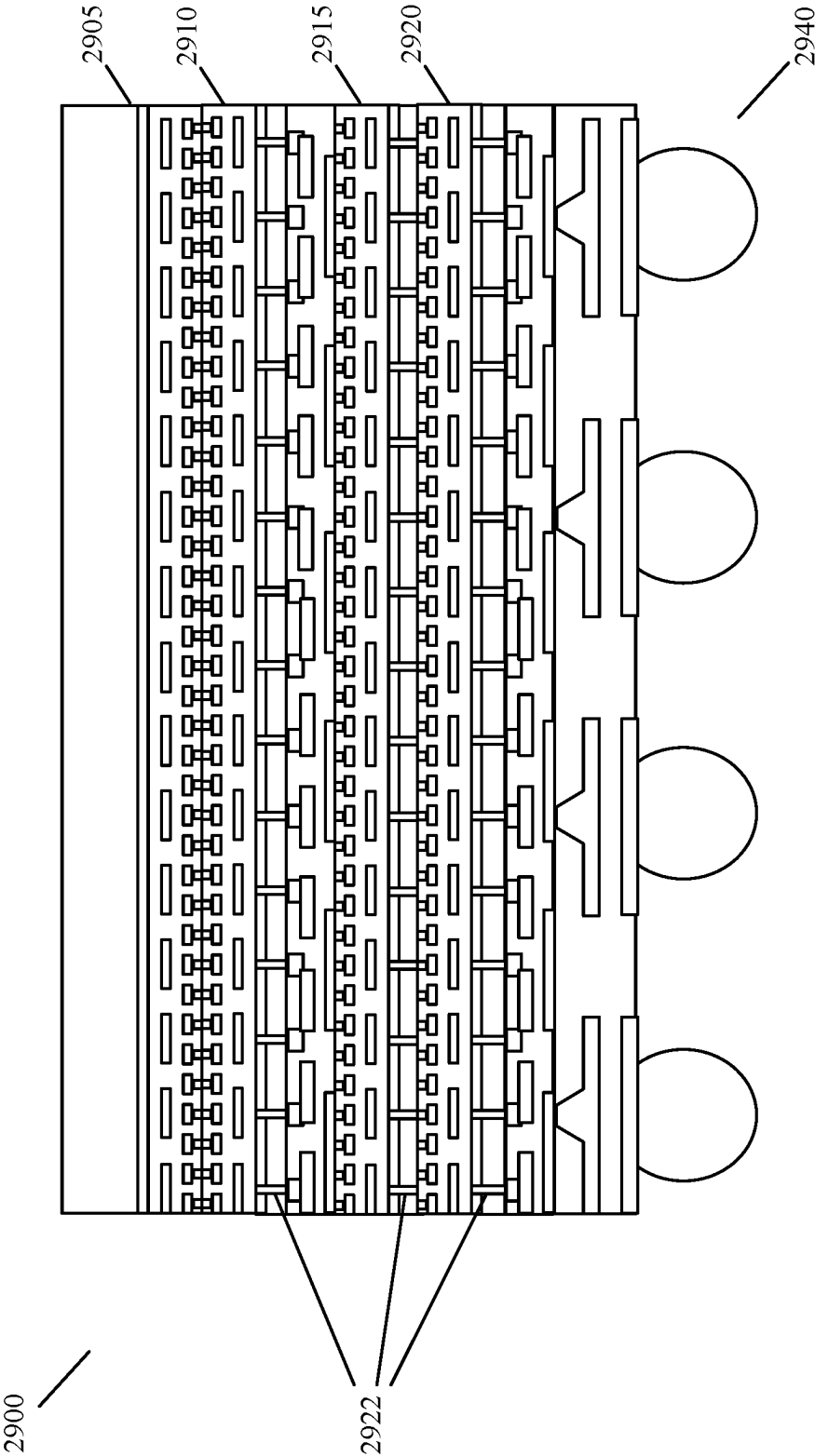


Figure 29

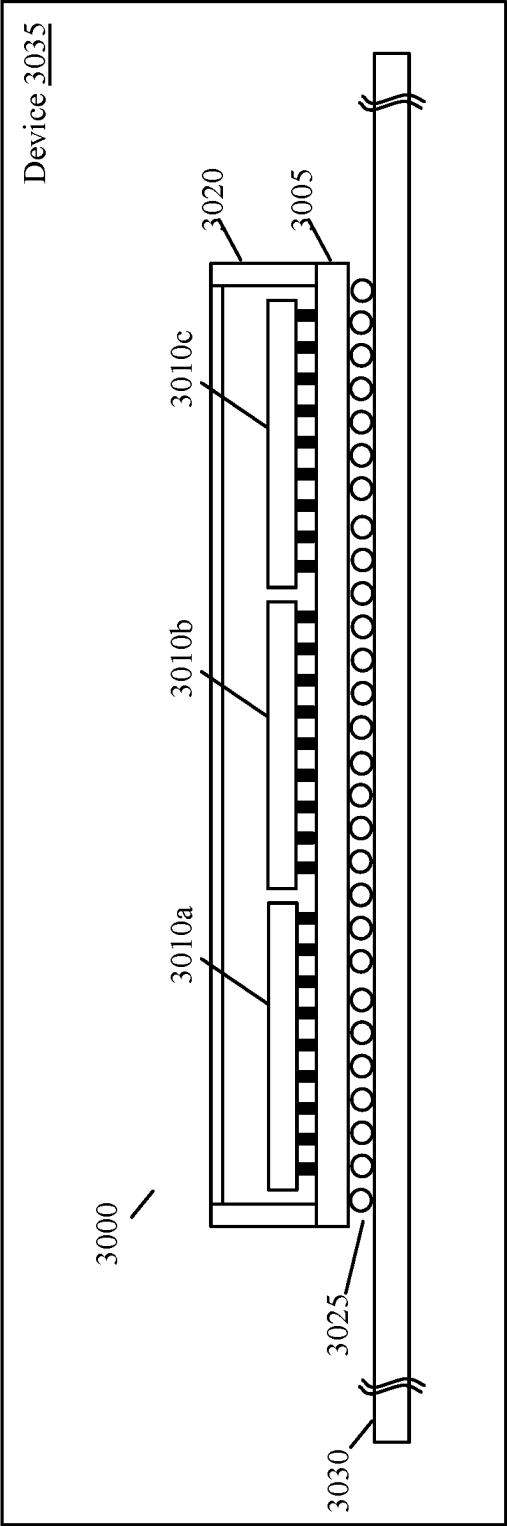


Figure 30

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3D PROCESSOR HAVING STACKED INTEGRATED CIRCUIT DIE

CLAIM OF BENEFIT

This application claims benefit to U.S. Provisional Patent Application 62/678,246 filed May 30, 2018, U.S. Provisional Patent Application 62/619,910 filed Jan. 21, 2018, U.S. Provisional Patent Application 62/575,221 filed Oct. 20, 2017, U.S. Provisional Patent Application 62/575,184 filed Oct. 20, 2017, U.S. Provisional Patent Application 62/575,240 filed Oct. 20, 2017, and U.S. Provisional Patent Application 62/575,259 filed Oct. 20, 2017. This application is a continuation-in-part of U.S. Non-Provisional patent application Ser. Nos. 15/859,546, 15/859,548, 15/859,551, and 15/859,612, all of which were filed on Dec. 31, 2017, and all of which claim benefit of U.S. Provisional Patent Application 62/541,064, filed on Aug. 3, 2017. This application is also a continuation-in-part of U.S. Non-Provisional patent application Ser. No. 15/976,809, filed on May 10, 2018, which claim benefit of U.S. Provisional Patent Application 62/619,910 filed Jan. 21, 2018. This application is also a continuation-in-part of U.S. Non-Provisional patent application Ser. No. 15/250,030, filed on Oct. 4, 2016, which claim benefit of U.S. Provisional Patent Application 62/405,833 filed Oct. 7, 2016. U.S. Provisional Patent Applications 62/678,246, 62/619,910, 62/575,221, 62/575,184, 62/575,240, 62/575,259, and 62/405,833 are incorporated herein by reference. U.S. Non-Provisional patent application Ser. Nos. 15/859,546, 15/859,548, 15/859,551, 15/859,612, 15/976,809, and 15/250,030 are incorporated herein by reference.

BACKGROUND

Electronic circuits are commonly fabricated on a wafer of semiconductor material, such as silicon. A wafer with such electronic circuits is typically cut into numerous dies, with each die being referred to as an integrated circuit (IC). Each die is housed in an IC case and is commonly referred to as a microchip, "chip," or IC chip. According to Moore's law (first proposed by Gordon Moore), the number of transistors that can be defined on an IC die will double approximately every two years. With advances in semiconductor fabrication processes, this law has held true for much of the past fifty years. However, in recent years, the end of Moore's law has been prognosticated as we are reaching the maximum number of transistors that can possibly be defined on a semiconductor substrate. Hence, there is a need in the art for other advances that would allow more transistors to be defined in an IC chip.

BRIEF SUMMARY

Some embodiments of the invention provide a three-dimensional (3D) circuit that is formed by vertically stacking two or more integrated circuit (IC) dies to at least partially overlap. In this arrangement, several circuit blocks defined on each die (1) overlap with other circuit blocks defined on one or more other dies, and (2) electrically connect to these other circuit blocks through connections that cross one or more bonding layers that bond one or more pairs of dies. The 3D circuit in some embodiments can be any type of circuit such as a processor, like a CPU (central processing unit), a GPU (graphics processing unit), a TPU (tensor processing unit), etc., or other kind of circuits, like

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an FPGA (field programmable gate array), AI (artificial intelligence) neural network chip, encrypting/decrypting chips, etc.

The connections in some embodiments cross the bonding layer(s) in a direction normal to the bonded surface. In some embodiments, the overlapping, connected circuit block pairs include pairs of computation blocks and pairs of computation and memory blocks. The connections that cross bonding layers to electrically connect circuit blocks on different dies are referred to below as z-axis wiring or connections. This is because these connections traverse completely or mostly in the z-axis of the 3D circuit, with the x-y axes of the 3D circuit defining the planar surface of the IC die substrate or interconnect layers. These connections are also referred to as vertical connections to differentiate them from the horizontal planar connections along the interconnect layers of the IC dies.

The preceding Summary is intended to serve as a brief introduction to some embodiments of the invention. It is not meant to be an introduction or overview of all inventive subject matter disclosed in this document. The Detailed Description that follows and the Drawings that are referred to in the Detailed Description will further describe the embodiments described in the Summary as well as other embodiments. Accordingly, to understand all the embodiments described by this document, a full review of the Summary, Detailed Description, the Drawings and the Claims is needed.

BRIEF DESCRIPTION OF THE DRAWINGS

The novel features of the invention are set forth in the appended claims. However, for purposes of explanation, several embodiments of the invention are set forth in the following figures.

FIG. 1 illustrates a 3D circuit of some embodiments of the invention.

FIG. 2 illustrates an example of a high-performance 3D processor that has a multi-core processor on one die and an embedded memory on another die.

FIG. 3 illustrates how multi-core processors are commonly used today in many devices.

FIG. 4 illustrates an example of a 3D processor that is formed by vertically stacking three dies.

FIG. 5 illustrates three vertically stacked dies with the backside of the second die thinned through a thinning process after face-to-face bonding the first and second dies but before face-to-back mounting the third die to the second die.

FIGS. 6-9 illustrate other 3D processors of some embodiments.

FIG. 10 illustrates that some embodiments place on different stacked dies two compute circuits that perform successive computations.

FIG. 11 illustrates an example of a high-performance 3D processor that has overlapping processor cores on different dies.

FIG. 12 illustrates another example of a high-performance 3D processor that has a processor core on one die overlap with a cache on another die.

FIG. 13 illustrates an example of a 3D processor that has different parts of a processor core on two face-to-face mounted dies.

FIG. 14 shows a compute circuit on a first die that overlaps a memory circuit on a second die, which is vertically stacked over the first die.

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FIG. 15 shows two overlapping compute circuits on two vertically stacked dies.

FIG. 16 illustrates an array of compute circuits on a first die overlapping an array of memories on a second die that is face-to-face mounted with the first die through direct bonded interconnect (DBI) bonding process.

FIG. 17 illustrates a traditional way of interlacing a memory array with a compute array.

FIGS. 18 and 19 illustrates two examples that show how high density DBI connections can be used to reduce the size of an arrangement of compute circuit that is formed by several successive stages of circuits, each of which performs a computation that produces a result that is passed to another stage of circuits until a final stage of circuits is reached.

FIG. 20 presents a compute circuit that performs a computation (e.g., an addition or multiplication) on sixteen multi-bit input values on two face-to-face mounted dies.

FIG. 21 illustrates a device that uses a 3D IC.

FIG. 22 provides an example of a 3D chip that is formed by two face-to-face mounted IC dies that are mounted on a ball grid array.

FIG. 23 illustrates a manufacturing process that some embodiments use to produce the 3D chip.

FIGS. 24-27 show two wafers at different stages of the fabrication process of FIG. 23.

FIG. 28 illustrates an example of a 3D chip with three stacked IC dies.

FIG. 29 illustrates an example of a 3D chip with four stacked IC dies.

FIG. 30 illustrates a 3D chip that is formed by face-to-face mounting three smaller dies on a larger die.

DETAILED DESCRIPTION

In the following detailed description of the invention, numerous details, examples, and embodiments of the invention are set forth and described. However, it will be clear and apparent to one skilled in the art that the invention is not limited to the embodiments set forth and that the invention may be practiced without some of the specific details and examples discussed.

Some embodiments of the invention provide a three-dimensional (3D) circuit that is formed by vertically stacking two or more integrated circuit (IC) dies to at least partially overlap. In this arrangement, several circuit blocks defined on each die (1) overlap with other circuit blocks defined on one or more other dies, and (2) electrically connect to these other circuit blocks through connections that cross one or more bonding layers that bond one or more pairs of dies. In some embodiments, the overlapping, connected circuit block pairs include pairs of computation blocks and pairs of computation and memory blocks.

In the discussion below, the connections that cross bonding layers to electrically connect circuit blocks on different dies are referred to below as z-axis wiring or connections. This is because these connections traverse completely or mostly in the z-axis of the 3D circuit (e.g., because these connections in some embodiments cross the bonding layer(s) in a direction normal or nearly normal to the bonded surface), with the x-y axes of the 3D circuit defining the planar surface of the IC die substrate or interconnect layers. These connections are also referred to as vertical connections to differentiate them from the horizontal planar connections along the interconnect layers of the IC dies.

The discussion above and below refers to different circuit blocks on different dies overlapping with each other. As illustrated in the figures described below, two circuit blocks

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on two vertically stacked dies overlap when their horizontal cross sections (i.e., their horizontal footprint) vertically overlap (i.e., have an overlap in the vertical direction).

FIG. 1 illustrates an example of such a 3D circuit. Specifically, it illustrates a 3D circuit 100 that is formed by vertically stacking two IC dies 105 and 110 such that each of several circuit blocks on one die (1) overlaps at least one other circuit block on the other die, and (2) electrically connects to the overlapping die in part through z-axis connections 150 that cross a bonding layer that bonds the two IC dies. In this example, the two dies 105 and 110 are face-to-face mounted as further described below. Also, although not shown in FIG. 1, the stacked first and second dies in some embodiments are encapsulated into one integrated circuit package by an encapsulating epoxy and/or a chip case.

As shown, the first die 105 includes a first semiconductor substrate 120 and a first set of interconnect layers 125 defined above the first semiconductor substrate 120. Similarly, the second IC die 110 includes a second semiconductor substrate 130 and a second set of interconnect layers 135 defined below the second semiconductor substrate 130. In some embodiments, numerous electronic components (e.g., active components, like transistors and diodes, or passive components, like resistors and capacitors) are defined on the first semiconductor substrate 120 and on the second semiconductor substrate 130.

The electronic components on the first substrate 120 are connected to each other through interconnect wiring on the first set of interconnect layers 125 to form numerous microcircuits (e.g., Boolean gates, such as AND gates, OR gates, etc.) and/or larger circuit blocks (e.g., functional blocks, such as memories, decoders, logic units, multipliers, adders, etc.). Similarly, the electronic components on the second substrate 130 are connected to each other through interconnect wiring on the second set of interconnect layers 135 to form additional microcircuits and/or larger circuit block.

In some embodiments, a portion of the interconnect wiring needed to define a circuit block on one die's substrate (e.g., substrate 120 of the first die 105) is provided by interconnect layer(s) (e.g., the second set interconnect layers 135) of the other die (e.g., the second die 110). In other words, the electronic components on one die's substrate (e.g., the first substrate 120 of the first die 105) in some embodiments are also connected to other electronic components on the same substrate (e.g., substrate 120) through interconnect wiring on the other die's set of interconnect layers (e.g., the second set of interconnect layers 135 of the second die 110) to form a circuit block on the first die.

As such, the interconnect layers of one die can be shared by the electronic components and circuits of the other die in some embodiments. The interconnect layers of one die can also be used to carry power, clock and data signals for the electronic components and circuits of the other die, as described in U.S. patent application Ser. No. 15/976,815 filed May 10, 2018, which is incorporated herein by reference. The interconnect layers that are shared between two dies are referred to as the shared interconnect layers in the discussion below.

Each interconnect layer of an IC die typically has a preferred wiring direction (also called routing direction). Also, in some embodiments, the preferred wiring directions of successive interconnect layers of an IC die are orthogonal to each other. For example, the preferred wiring directions of an IC die typically alternate between horizontal and vertical preferred wiring directions, although several wiring architectures have been introduced that employ 45 degree and 60

degree offset between the preferred wiring directions of successive interconnect layers. Alternating the wiring directions between successive interconnect layers of an IC die has several advantages, such as providing better signal routing and avoiding capacitive coupling between long parallel segments on adjacent interconnect layers.

To form the 3D circuit **100** of FIG. **1**, the first and second dies are face-to-face stacked so that the first and second set of interconnect layers **125** and **135** are facing each other. The top interconnect layers **160** and **165** are bonded to each other through a direct bonding process that establishes direct-contact metal-to-metal bonding, oxide bonding, or fusion bonding between these two sets of interconnect layers. An example of such bonding is copper-to-copper (Cu—Cu) metallic bonding between two copper conductors in direct contact. In some embodiments, the direct bonding is provided by a hybrid bonding technique such as DBI® (direct bond interconnect) technology, and other metal bonding techniques (such as those offered by Invenas Bonding Technologies, Inc., an Xperi Corporation company, San Jose, CA). In some embodiments, DBI connects span across silicon oxide and silicon nitride surfaces.

The DBI process is further described in U.S. Pat. Nos. 6,962,835 and 7,485,968, both of which are incorporated herein by reference. This process is also described in U.S. patent application Ser. No. 15/725,030, which is also incorporated herein by reference. As described in U.S. patent application Ser. No. 15/725,030, the direct bonded connections between two face-to-face mounted IC dies are native interconnects that allow signals to span two different dies with no standard interfaces and no input/output protocols at the cross-die boundaries. In other words, the direct bonded interconnects allow native signals from one die to pass directly to the other die with no modification of the native signal or negligible modification of the native signal, thereby forgoing standard interfacing and consortium-imposed input/output protocols.

Direct bonded interconnects allow circuits to be formed across and/or to be accessed through the cross-die boundary of two face-to-face mounted dies. Examples of such circuits are further described in U.S. patent application Ser. No. 15/725,030. The incorporated U.S. Pat. Nos. 6,962,835, 7,485,968, and U.S. patent application Ser. No. 15/725,030 also describe fabrication techniques for manufacturing two face-to-face mounted dies.

A DBI connection between two dies terminates on electrical contacts (referred to as pads in this document) on each die's top interconnect layer. Through interconnect lines and/or vias on each die, the DBI-connection pad on each die electrically connects the DBI connection with circuit nodes on the die that need to provide the signal to the DBI connection or to receive the signal from the DBI connection. For instance, a DBI-connection pad connects to an interconnect segment on the top interconnect layer of a die, which then carries the signal to a circuit block on the die's substrate through a series of vias and interconnect lines. Vias are z-axis structures on each die that carry signals between the interconnect layers of the die and between the IC die substrate and the interconnect layers of the die.

As shown in FIG. **1**, the direct bonding techniques of some embodiments allow a large number of direct connections **150** to be established between the top interconnect layer **165** of the second die **110** and top interconnect layer **160** of the first die **105**. For these signals to traverse to other interconnect layers of the first die **105** or to the substrate **120** of the first die **105**, the first die in some embodiments uses other IC structures (e.g., vias) to carry these signals from its

top interconnect layer to these other layers and/or substrate. In some embodiments, more than 1,000 connections/mm², 10,000 connections/mm², 100,000 connections/mm², 1,000,000 connections/mm² or less, etc. can be established between the top interconnect layers **160** and **165** of the first and second dies **105** and **110** in order to allow signals to traverse between the first and second IC dies.

The direct-bonded connections **150** between the first and second dies are very short in length. For instance, based on current manufacturing technologies, the direct-bonded connections can range from a fraction of a micron to a single-digit or low double-digit microns (e.g., 2-10 microns). As further described below, the short length of these connections allows the signals traversing through these connections to reach their destinations quickly while experiencing no or minimal capacitive load from nearby planar wiring and nearby direct-bonded vertical connections. The planar wiring connections are referred to as x-y wiring or connections, as such wiring stays mostly within a plane defined by an x-y axis of the 3D circuit. On the other hand, vertical connections between two dies or between two interconnect layers are referred to as z-axis wiring or connections, as such wiring mostly traverses in the z-axis of the 3D circuit. The use of "vertical" in expressing a z-axis connection should not be confused with horizontal or vertical preferred direction planar wiring that traverses an individual interconnect layer.

In some embodiments, the pitch (distance) between two neighboring direct-bonded connections **150** can be extremely small, e.g., the pitch for two neighboring connections is between 0.5 μm to 15 μm . This close proximity allows for the large number and high density of such connections between the top interconnect layers **160** and **165** of the first and second dies **105** and **110**. Moreover, the close proximity of these connections does not introduce much capacitive load between two neighboring z-axis connections because of their short length and small interconnect pad size. For instance, in some embodiments, the direct bonded connections are less than 1 or 2 μm in length (e.g., 0.1 to 0.5 μm in length), and facilitate short z-axis connections (e.g., 1 to 10 μm in length) between two different locations on the two dies even after accounting for the length of vias on each of the dies. In sum, the direct vertical connections between two dies offer short, fast paths between different locations on these dies.

Through the z-axis connections **150** (e.g., DBI connections), electrical nodes in overlapping portions of the circuit blocks on the first and second dies can be electrically connected. These electrical nodes can be on the IC die substrates (e.g., on the portions of the substrates that contain node of electronic components of the circuit blocks) or on the IC die interconnect layers (e.g., on the interconnect layer wiring that form the circuit block). When these electrical nodes are not on the top interconnect layers that are connected through the z-axis connections, vias are used to carry the signals to or from the z-axis connections to these nodes. On each IC die, vias are z-axis structures that carry signals between the interconnect layers and between the IC die substrate and the interconnect layers.

FIG. **1** illustrates numerous z-axis connections **150** between overlapping regions **181-185** in the top interconnect layers **160** and **165**. Each of these regions corresponds to a circuit block **171-175** that is defined on one of the IC die substrates **120** and **130**. Also, each region on the top interconnect layer of one die connects to one or more overlapping regions in the top interconnect layer of the other die through numerous z-axis connections. Specifically, as shown, z-axis

connections connect overlapping regions **181** and **184**, regions **182** and **184**, and regions **183** and **185**. Vias are used to provide signals to these z-axis connections from the IC die substrates and interconnect layers. Also, vias are used to carry signals from the z-axis connections when the electrical nodes that need to receive these signals are on the die substrates or the interconnect layers below the top layer.

When the z-axis connections are DBI connections, the density of connections between overlapping connected regions can be in the range of 1,000 connections/mm² to 1,000,000 connections/mm². Also, the pitch between two neighboring direct-bonded connections **150** can be extremely small, e.g., the pitch for two neighboring connections is between 0.5 μ m to 15 μ m. In addition, these connections can be very short, e.g., in the range from a fraction of a micron to a low single-digit microns. These short DBI connections would allow very short signal paths (e.g., single digit or low-double digit microns, such as 2-20 microns) between two electrically connected circuit nodes on the two substrates of the IC dies **105** and **110** even after accounting for interconnect-layer vias and wires.

In the example illustrated in FIG. 1, each top interconnect-layer region **181-185** corresponds to a circuit block region **171-175** on an IC die substrate **120** or **130**. One of ordinary skill will realize that a circuit block's corresponding top interconnect-layer region (i.e., the region that is used to establish the z-axis connections for that circuit block) does not have to perfectly overlap the circuit block's region on the IC substrate. Moreover, in some embodiments, all the z-axis connections that are used to connect two overlapping circuit blocks in two different dies do not connect one contiguous region in the top-interconnect layer of one die with another contiguous region in the top-interconnect layer of the other die.

Also, in some embodiments, the z-axis connections connect circuits on the two dies that do not overlap (i.e., do not have any of their horizontal cross section vertically overlap). However, it is beneficial to use z-axis connections to electrically connect overlapping circuits (e.g., circuit blocks **173** and **175**, circuit blocks **171** and **174**, etc.) on the two dies **105** and **110** (i.e., circuits with horizontal cross sections that vertically overlap) because such overlaps dramatically increase the number of candidate locations for connecting the two circuits. When two circuits are placed next to each other on one substrate, the number of connections that can be established between them is limited by the number of connections that can be made through their perimeters on one or more interconnect layers. However, by placing the two circuits in two overlapping regions on two vertically stacked dies, the connections between the two circuits are not limited to periphery connections that come through the perimeter of the circuits, but also include z-axis connections (e.g., DBI connections and via connections) that are available through the area of the overlapping region.

Stacking IC dies in many cases allows the wiring for delivering the signals to be much shorter, as the stacking provides more candidate locations for shorter connections between overlapping circuit blocks that need to be interconnected to receive these signals. For instance, in FIG. 1, circuit blocks **173** and **175** on dies **105** and **110** share a data bus **190** on the top interconnect layer of the second die **110**. This data bus carries data signals to both of these circuits.

Direct-bonded connections are used to carry signals from this data bus **190** to the circuit block **175** on the first die **105**. These direct-bonded connections are much shorter than connections that would route data-bus signals on the first die about several functional blocks in order to reach the circuit

block **175** from this block's periphery. The data signals that traverse the short direct-bonded connections reach this circuit **175** on the first die very quickly (e.g., within 1 or 2 clock cycles) as they do not need to be routed from the periphery of the destination block. On a less-congested shared interconnect layer, a data-bus line can be positioned over or near a destination circuit on the first die to ensure that the data-bus signal on this line can be provided to the destination circuit through a short direct-bonded connection.

Z-axis connection and the ability to share interconnect layers on multiple dies reduce the congestion and route limitations that may be more constrained on one die than another. Stacking IC dies also reduces the overall number of interconnect layers of the two dies because it allows the two dies to share some of the higher-level interconnect layers in order to distribute signals. Reducing the higher-level interconnect layers is beneficial as the wiring on these layers often consumes more space due to their thicker, wider and coarser arrangements.

Even though in FIG. 1 the two dies are face-to-face mounted, one of ordinary skill will realize that in other embodiments two dies are vertically stacked in other arrangements. For instance, in some embodiments, these two dies are face-to-back stacked (i.e., the set of interconnect layers of one die is mounted next to the backside of the semiconductor substrate of the other die), or back-to-back stacked (i.e., the backside of the semiconductor substrate of one die is mounted next to the backside of the semiconductor substrate of the other die).

In other embodiments, a third die (e.g., an interposer die) is placed between the first and second dies, which are face-to-face stacked, face-to-back stacked (with the third die between the backside of the substrate of one die and the set of interconnect layers of the other die), or back-to-back stacked (with the third die between the backsides of the substrates of the first and second dies). Also, as further described by reference to FIG. 4, the vertical stack of dies in some embodiments includes three or more IC dies in a stack. While some embodiments use a direct bonding technique to establish connections between the top interconnect layers of two face-to-face stacked dies, other embodiments use alternative connection schemes (such as through silicon vias, TSVs, through-oxide vias, TOVs, or through-glass vias, TGVs) to establish connections between face-to-back dies and between back-to-back dies.

In FIG. 1, the overlapping circuit blocks **171-175** on the two dies **105** and **110** are different types of blocks in different embodiments. Examples of such blocks in some embodiments include memory blocks that store data, computational blocks that perform computations on the data, and I/O blocks that receive and output data from the 3D circuit **100**. To provide more specific examples of overlapping circuit blocks, FIGS. 2, 4, and 6 illustrate several different overlapping memory blocks, computational blocks, and/or I/O blocks architectures of some embodiments. Some of these examples illustrate high performance 3D multi-core processors. FIGS. 10-11 then illustrate several examples of overlapping computation blocks, including different cores of a multi-core processor being placed on different IC dies. FIG. 13 illustrates an example of overlapping functional blocks of a processor core.

FIG. 2 illustrates an example of a high-performance 3D processor **200** that has a multi-core processor **250** on one die **205** and an embedded memory **255** on another die **210**. As shown in this figure, the horizontal cross section of the multi-core processor has a substantially vertical overlaps with the horizontal cross section of the embedded memory.

Also, in this example, the two dies **205** and **210** are face-to-face mounted through a direct bonding process, such as the DBI process. In other embodiments, these two dies can be face-to-back or back-to-back mounted.

As shown in FIG. 2, numerous z-axis connections **150** cross a direct bonding layer that bonds the two IC dies **205** and **210** in order to establish numerous signal paths between the multi-core processor **250** and the embedded memory **255**. When the DBI process is used to bond the two dies **205** and **210**, the z-axis connections can be in the range of 1,000 connections/mm² to 1,000,000 connections/mm². As such, the DBI z-axis connections allow a very large number of signal paths to be defined between the multi-core processor **250** and the embedded memory **255**.

The DBI z-axis connections **150** also support very fast signal paths as the DBI connections are typically very short (e.g., are 0.2 μm to 2 μm). The overall length of the signal paths is also typically short because the signal paths are mostly vertical. The signal paths often rely on interconnect lines (on the interconnect layers) and vias (between the interconnect layers) to connect nodes of the processor **250** and the embedded memory **255**. However, the signal paths are mostly vertical as they often connect nodes that are in the same proximate z-cross section. Given that the DBI connections are very short, the length of a vertical signal path mostly accounts for the height of the interconnect layers of the dies **205** and **210**, which is typically in the single digit to low-double digit microns (e.g., the vertical signal paths are typically in the range of 10-20 μm long).

As z-axis connections provide short, fast and plentiful connections between the multi-core processor **250** and the embedded memory **255**, they allow the embedded memory **255** to replace many of the external memories that are commonly used today in devices that employ multi-core processors. In other words, robust z-axis connections between vertically stacked IC dies enable next generation system on chip (SoC) architectures that combine the computational power of the fastest multi-core processors with large embedded memories that take the place of external memories.

To better illustrate this, FIG. 3 illustrates how multi-core processors are commonly used today in many devices. As shown, a multi-core processor **350** in a device **305** typically communicates with multiple external memories **310** of the device **305** through an external I/O interface **355** (such as a double data rate (DDR) interface). As further show, the multi-core processor has multiple general processing cores **352** and one or more graphical processing cores **354** that form a graphical processing unit **356** of the processor **350**.

Each of the processing cores has its own level 1 (L1) cache **362** to store data. Also, multiple level 2 (L2) caches **364** are used to allow different processing cores to store their data for access by themselves and by other cores. One or more level 3 (L3) caches **366** are also used to store data retrieved from external memories **310** and to supply data to external memories **310**. The different cores access the L2 and L3 caches through arbiters **368**. As shown, I/O interfaces **355** are used to retrieve data for L3 cache **366** and the processing cores **352** and **354**. L1 caches typically have faster access times than L2 caches, which, in turn, have faster access times often than L3 caches.

The I/O interfaces consume a lot of power and also have limited I/O capabilities. Often, I/O interfaces have to serialize and de-serialize the output data and the input data, which consumes power and also restricts the multi-core processors input/output. Also, the architecture illustrated in

FIG. 3 requires enough wiring to route the signals between the various components of the multi-core processor and the I/O interfaces.

The power consumption, wiring and processor's I/O bottleneck is dramatically improved by replacing the external memories with one or more embedded memories **255** that are vertically stacked with the multi-core processor **250** in the same IC package. This arrangement dramatically reduces the length of the wires needed to carry signals between the multi-core processor **250** and its external memory (which in FIG. 2 is the embedded memory **255**). Instead of being millimeters in length, this wiring is now in the low microns. This is a 100-1000 times improvement in wirelength.

The reduction in wirelength allows the 3D processor **200** of FIG. 2 to have much lower power consumption than the present day design of FIG. 3. The 3D processor's stacked design also consumes much less power as it foregoes the low throughput, high power consuming I/O interface between the external memories **310** and the multi-core processor **350** with plentiful, short z-axis connections between the embedded memory **255** and the multi-core processor **250**. The 3D processor **200** still needs an I/O interface on one of its dies (e.g., the first die **205**, the second die **210** or another stacked die, not shown), but this processor **200** does not need to rely on it as heavily to input data for consumption as a large amount of data (e.g., more than 200 MB, 500 MB, 1 GB, etc.) can be stored in the embedded memory **255**.

The stacked design of the 3D processor **200**, FIG. 2, also reduces the size of the multi-core processor by requiring less I/O interface circuits and by placing the I/O interface circuits **257** on the second die **210**. In other embodiments, the I/O interface circuits **257** are on the first die **205**, but are fewer and/or smaller circuits. In still other embodiments, the I/O interface circuits are placed on a third die stacked with the first and second dies, as further described below.

The stacked design of the 3D processor **200** also frees up space in the device that uses the multi-core processor as it moves some of the external memories to be in the same IC chip housing as the multi-core processor. Examples of memories that can be embedded memories **255** stacked with the multi-core processor **250** include any type of memory, such as SRAM (static random access memory), DRAM (dynamic random access memory), MRAM (magnetoresistive random access memory), TCAM (ternary content addressable random access memory), NAND Flash, NOR Flash, RRAM (resistive random access memory), PCRAM (phase change random access memory), etc.

Even though FIG. 2 illustrates one embedded memory on the second die **210**, multiple embedded memories are defined on the second die **210** in some embodiments, while multiple embedded memories are defined on two or more dies that are vertically stacked with the first die **205** that contains the multi-core processor **250**. In some embodiments that use multiple different embedded memories, the different embedded memories all are of the same type, while in other embodiments, the different embedded memories are different types (e.g., some are SRAMs while others are NAND/NOR Flash memories). In some embodiments, the different embedded memories are defined on the same IC die, while in other embodiments, different embedded memories are defined on different IC dies.

FIG. 2 illustrates that in some embodiments the multi-core processor **250** has the similar components (e.g., multiple general processing cores **270**, L1, L2, and L3 caches **272**, **274**, and **276**, cache arbiters **278** and **280**, graphical processing core **282**, etc.) like other multi-processor cores.

However, in the 3D processor **200**, the I/O interface circuits **257** for the multi-core processor **250** are placed on the second die **205**, as mentioned above.

The I/O circuits **257** write data to the embedded memory **255** from external devices and memories, and reads data from the embedded memory **255** for the external devices and memories. In some embodiments, the I/O circuit **255** can also retrieve data from external devices and memories for the L3 cache, or receive data from the L3 cache for external devices and memories, without the data first going through the embedded memory **255**. Some of these embodiments have a direct vertical (z-axis) bus between the L3 cache and the I/O circuit **257**. In these or other embodiments, the first die **205** also includes I/O circuits as interfaces between the I/O circuit **255** and the L3 cache **276**, or as interfaces between the L3 cache **276** and the external devices/memories.

Instead of, or in conjunction with, placing I/O circuits on a different die than the rest of the multi-core processor, some embodiments place other components of a multi-core processor on different IC dies that are placed in a vertical stack. For instance, FIG. 4 illustrates an example of a 3D processor **400** that is formed by vertically stacking three dies **405**, **410** and **415**, with the first die **405** including multiple processor cores **422** and **424** of a multi-core processor, the second die **410** including L1-L3 caches **426**, **428** and **430** for the processing cores, and the third die **415** including I/O circuits **435**. In this example, the first and second dies **405** and **410** are face-to-face mounted (e.g., through a direct bonding process, such as a DBI process), while the second and third dies **410** and **415** are back-to-face mounted.

In this example, the processor cores are in two sets of four cores **432** and **434**. As shown, each core on the first die **405** overlaps (1) with that core's L1 cache **426** on the second die **410**, (2) with one L2 cache **428** on the second die **410** that is shared by the three other cores in the same four-core set **432** or **434**, and (3) with the L3 cache **430** on the second die **410**. In some embodiments, numerous z-axis connections (e.g., DBI connections) establish numerous signal paths between each core and each L1, L2, or L3 cache that it overlaps. These signal paths are also established by interconnect segments on the interconnect layers, and vias between the interconnect layers, of the first and second dies.

In some embodiments, some or all of the cache memories (e.g., the L2 and L3 caches **428** and **430**) are multi-ported memories that can be simultaneously accessed by different cores. One or more of the cache memories in some embodiments include cache arbiter circuits that arbitrate (e.g., control and regulate) simultaneous and at time conflicting access to the memories by different processing cores. As shown, the 3D processor **400** also includes one L2 cache memory **436** on the first die **405** between the two four-core sets **432** and **434** in order to allow data to be shared between these sets of processor cores. In some embodiments, the L2 cache memory **436** includes a cache arbiter circuit (not shown). In other embodiments, the 3D processor **400** does not include the L2 cache memory **436**. In some of these embodiments, the different processor core sets **432** and **434** share data through the L3 cache **430**.

The L3 cache **430** stores data for all processing cores **422** and **424** to access. Some of this data is retrieved from external memories (i.e., memories outside of the 3D processor **400**) by the I/O circuit **435** that is defined on the third die **415**. The third die **415** in some embodiments is face-to-back mounted with the second die. To establish this mounting, TSVs **460** are defined through the second die's substrate, and these TSVs electrically connect (either directly or

through interconnect segments defined on the back side of the second die) to direct bonded connections that connect the backside of the second die to the front side of the third die (i.e., to the top interconnect layer on the front side of the third die). As shown in FIG. 5, the backside of the second die is thinned through a thinning process after face-to-face bonding the first and second dies but before face-to-back mounting the third die to the second die. This thinning allows the TSVs through the second die's substrate to be shorter. The shorter length of the TSVs, in turn, allows the TSVs to have smaller cross sections and smaller pitch (i.e., smaller center-to-center distance to neighboring TSVs), which thereby improves their density.

Most of the signal paths between the second and third dies **410** and **415** are very short (e.g., typically in the range of 10-20 μm long) as they mostly traverse in the vertical direction through the thinned second die's substrate and third die's interconnect layers, which have relatively short heights. In some embodiments, a large number of short, vertical signal paths are defined between the L3 cache **430** on the second die **410** and the I/O circuit **435** on the third die **415**. These signal paths use (1) direct-bonded connections between the top interconnect layer of the third die **415** and the backside of the second die **410**, (2) TSVs **460** through the second die's substrate, and (3) vias between the interconnect layers, and interconnect segments on the interconnect layers, of the second and third die. The number and short length of these signal paths allow the I/O circuit to rapidly write to and read from the L3 cache.

The signal paths between the first and second dies **405** and **410** use (1) direct-bonded connections between the top interconnect layers of the first and second dies **405** and **410**, and (2) vias between the interconnect layers, and interconnect segments on the interconnect layers, of the first and second dies **405** and **410**. Most of these signal paths between the first and second dies **405** and **410** are also very short (e.g., typically in the range of 10-20 μm long) as they mostly traverse in the vertical direction through the first and second dies' interconnect layers, which have relatively short heights. In some embodiments, a large number of short, vertical signal paths are defined between the processing cores on the first die **405** and their associated L1-L3 caches.

In some embodiments, the processor cores use these fast and plentiful signal paths to perform very fast writes and reads of large data bit sets to and from the L1-L3 cache memories. The processor cores then perform their operations (e.g., their instruction fetch, instruction decode, arithmetic logic, and data write back operations) based on these larger data sets, which in turn allows them to perform more complex instruction sets and/or to perform smaller instruction sets more quickly.

FIG. 6 illustrates another 3D processor **600** of some embodiments. This processor **600** combines features of the 3D processor **200** of FIG. 2 with features of the 3D processor **400** of FIG. 4. Specifically, like the processor **400**, the processor **600** places multiple processor cores **422** and **424** on a first die **605**, L1-L3 caches **426**, **428** and **430** on a second die **610**, and I/O circuits **435** on a third die **615**. However, like the processor **200**, the processor **600** also has one die with an embedded memory **622**. This embedded memory is defined on a fourth die **620** that is placed between the second and third dies **610** and **615**.

In FIG. 6, the first and second dies **605** and **610** are face-to-face mounted (e.g., through a direct bonding process, such as a DBI process), the fourth and second dies **620** and **610** are face-to-back mounted, and the third and fourth dies **615** and **620** are face-to-back mounted. To establish the

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face-to-back mounting, TSVs **460** are defined through the substrates of the second die and third dies. The TSVs through the second die **610** electrically connect (either directly or through interconnect segments defined on the back side of the second die) to direct bonded connections that connect the backside of the second die **610** to the front side of the fourth die **620**, while the TSVs through the fourth die **620** electrically connect (either directly or through interconnect segments defined on the back side of the second die) to direct bonded connections that connect the backside of the fourth die **620** to the front side of the third die **615**.

To allow these TSVs to be shorter, the backside of the second die is thinned through a thinning process after face-to-face bonding the first and second dies but before face-to-back mounting the fourth die **620** to the second die **610**. Similarly, the backside of the fourth die **620** is thinned through a thinning process after face-to-back mounting the fourth and second dies **620** and **610** but before face-to-back mounting the third die **615** to the fourth die **620**. Again, the shorter length of the TSVs allows the TSVs to have smaller cross sections and smaller pitch (i.e., smaller center-to-center distance to neighboring TSVs), which thereby improves their density.

As in FIG. 4, the L3 cache **430** in FIG. 6 stores data for all processing cores **422** and **424** to access. However, in FIG. 6, the L3 cache does not connect to the I/O circuits **435** but rather connects to the embedded memory **622** on the fourth die through vertical signal paths. In this design, the embedded memory **622** connects to the I/O circuits **435** on the third die **615** through vertical signal paths. In some embodiments, the vertical signal paths between the second and fourth dies **610** and **620** and between the fourth and third dies **620** and **615** are established by z-axis direct bonded connections and TSVs, as well as interconnect segments on the interconnect layers and vias between the interconnect layers. Most of these signal paths are very short (e.g., typically in the range of 10-20 μm long) as they are mostly vertical and the height of the thinned substrates and their associated interconnect layers is relatively short.

Like the embedded memory **255** of FIG. 2, the embedded memory **622** is a large memory (e.g., is larger than 200 MB, 500 MB, 1 GB, etc.) in some embodiments. As such, the embedded memory in some embodiments can replace one or more external memories that are commonly used today in devices that employ multi-core processors. Examples of the embedded memory **622** include SRAM, DRAM, MRAM, NAND Flash, NOR Flash, RRAM, PCRAM, etc. In some embodiments, two or more different types of embedded memories are defined on one die or multiple dies in the stack of dies that includes one or more dies on which a multi-core processor is defined.

Through numerous short, vertical signal paths, the embedded memory **622** receives data from, and supplies data to, the I/O circuit **435**. Through these signal paths, the I/O circuit **435** writes data to the embedded memory **622** from external devices and memories, and reads data from the embedded memory **622** for the external devices and memories. In some embodiments, the I/O circuit **435** can also retrieve data from external devices and memories for the L3 cache, or receive data from the L3 cache for external devices and memories, without the data first going through the embedded memory **622**. Some of these embodiments have a direct vertical (z-axis) bus between the L3 cache and the I/O circuit **435**. In these or other embodiments, the second die **610** and/or fourth die **620** also include I/O circuits as

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interfaces between the I/O circuit **435** and the L3 cache **430**, or as interfaces between the L3 cache **430** and the external devices/memories.

FIG. 7 illustrates yet another 3D processor **700** of some embodiments. This processor **700** is identical to the processor **600** of FIG. 6, except that it only has two layers of caches, L1 and L2, on a second die **710** that is face-to-face mounted on a first die **705** that has eight processor cores **722**. As shown, each L1 cache **726** overlaps just one core **722**. Unlike the L1 caches **726**, the L2 cache **728** is shared among all the cores **722** and overlaps each of the cores **722**. In some embodiments, each core connects to each L1 or L2 cache that it overlaps through (1) numerous z-axis DBI connections that connect the top interconnect layers of the dies **705** and **721**, and (2) the interconnects and vias that carry the signals from these DBI connections to other metal and substrate layers of the dies **705** and **710**. The DBI connections in some embodiments allow the data buses between the caches and the cores to be much wider and faster than traditional data buses between the caches and the cores.

In some embodiments, L1 caches are formed by memories that can be accessed faster (i.e., have faster read or write times) than the memories that are used to form L2 caches. Each L1 cache **726** in some embodiments is composed of just one bank of memories, while in other embodiments it is composed of several banks of memories. Similarly, the L2 cache **728** in some embodiments is composed of just one bank of memories, while in other embodiments it is composed of several banks of memories. Also, in some embodiments, the L1 caches **726** and/or L2 cache **728** are denser than traditional L1 and L2 caches as they use z-axis DBI connections to provide and receive their signals to and from the overlapping cores **722**. In some embodiments, the L1 and L2 caches **726** and **728** are much larger than traditional L1 and L2 caches as they are defined on another die than the die on which the cores are defined, and hence face less space restrictions on their placement and the amount of space that they consume on the chip.

Other embodiments use still other architectures for 3D processors. For example, instead of using just one L2 cache **728**, some embodiments use two or four L2 caches that overlap four cores (e.g., the four left cores **726** and the four right cores **726**) or two cores (e.g., one of the four pairs of vertically aligned cores **722**). FIG. 8 illustrates another 3D processor **800** of some embodiments. This processor **800** is identical to the processor **700** of FIG. 7, except that it does not have the L2 cache **728**. In place of this L2 cache, the processor **900** has a network on chip (NOC) **8028** on the die **810**, which is face-to-face mounted to the die **705** through a DBI bonding process.

In some embodiments, the NOC **828** is an interface through which the cores **722** communicate. This interface includes one or more buses and associated bus circuitry. The NOC **828** in some embodiments also communicatively connects each core to the L1 caches that overlap the other cores. Through this NOC, a first core can access data stored by a second core in the L1 cache that overlap the second core. Also, through this NOC, a first core in some embodiments can store data in the L1 cache that overlaps a second core. In some embodiments, an L1 and L2 cache overlaps each core **722**, and the NOC **828** connects the cores to L2 caches of other cores, but not to the L1 caches of these cores. In other embodiments, the NOC **828** connects the cores to both L1 and L2 caches that overlap other cores, as well as to the other cores.

FIG. 9 illustrates yet another 3D processor **900** of some embodiments. This processor **900** is identical to the proces-

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sor **400** of FIG. 4, except that it only has one L1 cache **932** on a die **910** for each of six CPU (central processing unit) cores **922** and one L1 cache **934** for each of two GPU (graphical processing unit) cores **924** that are defined on a die **905** that is face-to-face mounted to the die **910** through a DBI bonding process. The processor **900** does not use layers 2 and 3 caches as it uses large L1 caches for its CPU and GPU cores. The L1 caches can be larger than traditional L1 caches as they are defined on another die than the die on which the cores are defined, and hence face less space restrictions on their placement and the amount of space that they consume on the chip.

In FIG. 9, the processor **900** has its I/O interface defined on a third die **415** that is face-to-back mounted on the die **910**. In other embodiments, the processor **900** does not include the third die **415**, but just includes the first and second dies **905** and **910**. In some of these embodiments, the I/O interface of the processor **900** is defined on the first and/or second dies **905** and **910**. Also, in other embodiments, one L1 cache **932** is shared across multiple CPU cores **922** and/or multiple GPU cores **924**.

FIG. 10 illustrates that some embodiments place on different stacked dies two compute circuits that perform successive computations. A compute circuit is a circuit that receives a multi-bit value as input and computes a multi-bit value as output based on the received input. In FIG. 10, one compute circuit **1015** is defined on a first die **1005** while the other compute circuit **1020** is defined on a second die **1010**.

The first and second dies are face-to-face mounted through a direct bonding process (e.g., a DBI process). This mounting defines numerous z-axis connections between the two dies **1005** and **1010**. Along with interconnect line on the interconnect layers, and vias between the interconnect layers, of the two dies, the z-axis connections define numerous vertical signal paths between the two compute circuits **1015** and **1020**. These vertical signal paths are short as they mostly traverse in the vertical direction through the die interconnect layers, which are relatively short. As they are very short, these vertical signal paths are very fast parallel paths that connect the two compute circuit **1015** and **1020**.

In FIG. 10, the first compute circuit **1015** receives a multi-bit input value **1030** and computes a multi-bit output value **1040** based on this input value. In some embodiments, the multi-bit input value **1030** and/or output value **1040** are large bit values, e.g., 32 bits, 64 bits, 128 bits, 256 bits, 512 bits, 1024 bits, etc. Through the vertical signal paths between these two compute circuits, the first compute circuit **1015** provides its multi-bit output value **1040** as the input value to the compute circuit **1020**. Based on this value, the compute circuit **1020** computes another multi-bit output value **1045**.

Given the large number of vertical signal paths between the first and second compute circuits **1015** and **1020**, large number of bits can be transferred between these two circuits **1015** and **1020** without the need to use serializing and de-serializing circuits. The number of the vertical signal paths and the size of the exchanged data also allow many more computations to be performed per each clock cycle. Because of the short length of these vertical signal paths, the two circuits **1015** and **1020** can exchange data within one clock cycle. When two computation circuits are placed on one die, it sometimes can take 8 or more clock cycles for signals to be provided from one circuit to another because of the distances and/or the congestion between the two circuits.

In some embodiments, the two overlapping computation circuits on the two dies **1005** and **1010** are different cores of a multi-core processor. FIG. 11 illustrates an example of a

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high-performance 3D processor **1100** that has overlapping processor cores on different dies. In this example, two dies **1105** and **1110** are face-to-face mounted through a direct bonding process (e.g., the DBI process). The first die **1105** includes a first processor core **1112**, while the second die **1110** includes a second processor core **1114**.

The first die **1105** also includes an L1 cache **1116** for the second core **1114** on the second die **1110**, and L2 and L3 caches **1122** and **1126** for both cores **1112** and **1114**. Similarly, the second die **1110** also includes an L1 cache **1118** for the first core **1112** on the first die **1105**, and L2 and L3 caches **1124** and **1128** for both cores **1112** and **1114**. As shown, each core completely overlaps its corresponding L1 cache, and connects to its L1 cache through numerous vertical signal paths that are partially defined by z-axis connections between the top two interconnect layers of the dies **1105** and **1110**. As mentioned above, such vertical signal paths are also defined by (1) vias between interconnect layers of each die, and/or (3) interconnect segments on interconnect layers of each die.

Each core on one die also overlaps with one L2 cache and one L3 cache on the other die and is positioned near another L2 cache and another L3 cache on its own die. Each L2 and L3 cache **1122-826** can be accessed by each core **1112** or **1114**. Each core accesses an overlapping L2 or L3 cache through numerous vertical signal paths that are partially defined by z-axis connections between the top two interconnect layers of the dies **1105** and **1110** and by (1) vias between interconnect layers of each die, and/or (3) interconnect segments on interconnect layers of each die.

Each core can also access an L2 or L3 cache on its own die through signal paths that are defined by vias between interconnect layers, and interconnect segments on interconnect layers, of its own die. In some embodiments, when additional signal paths are needed between each core and an L2 or L3 cache on its own die, each core also connects to such L2 or L3 cache through signal paths that are not only defined by vias between interconnect layers, and interconnect segments on interconnect layers, of its own die, but by vias between interconnect layers and interconnect segments on interconnect layers of the other die.

Other embodiments, however, do not use signal paths that traverse through the other die's interconnect layers to connect a core with an L2 cache or an L3 cache on its own die, because such signal paths might have different delay (i.e., a greater delay) than signal paths between this core and this cache that only use the interconnect layers of the core's own die. On the other hand, given the very short length of the z-axis connections, other embodiments use signal paths defined through the other die's interconnect layers (e.g., through its top interconnect layers) when the difference in the signal path delay is very small (as compared to the speed of the signal paths that only use the interconnect layers of the core's die).

The 3D architecture illustrated in FIG. 11 dramatically increases the number of connections (through vertical signal paths) between each core **1112** or **1114** and its corresponding L1, L2 and L3 caches. With this increase, each core **1112** or **1114** retrieves much larger sets of data bits and performs more complex operations faster with such larger sets of data bits. In some embodiments, each core uses wider instruction and data buses in its pipelines as it can retrieve wider instructions and data from overlapping memories. In these or other embodiments, each core has more pipelines that perform more operations in parallel as the core can retrieve more instruction and data bits from the overlapping memories.

In some embodiments, each core on one die only uses the L2 cache or L3 cache on the other die (i.e., only uses the L2 or L3 cache that vertically overlaps the core) in order to take advantage of the large number of vertical signal paths between it and the overlapping L2 cache. Each core in some of these embodiments stores a redundant copy of each data, which it stores in its own overlapping cache (e.g., its own overlapping L2 cache) in the corresponding cache (e.g., in the other L2 cache) that is defined on the core's own die, so that the data is also available for the other core. In some of these embodiments, each core reaches the cache on its own die through signal paths that are not only defined through the interconnect lines and vias on the core's die, but also defined through interconnect lines and vias of the other die.

FIG. 12 illustrates another example of a high-performance 3D processor **1200** that has a processor core on one die overlap with a cache on another die. In this example, two dies **1205** and **1210** are face-to-face mounted through a direct bonding process (e.g., the DBI process). The first die **1205** includes a first processor core **1212**, while the second die **1210** includes a second processor core **1214**. The first die **1205** includes an L1 cache **1216** for the second processor core **1214** defined on the second die **1210**, while the second die **1210** includes an L1 cache **1218** for the first processor core **1212** defined on the first die **1205**.

In this example, the cross-section of each L1 cache on one die completely overlaps the cross-section of the corresponding core on the other die. This ensures the largest region for defining z-axis connections (e.g., DBI connections) in the overlapping regions of each core and its corresponding L1 cache. These z-axis connections are very short and hence can be used to define a very fast bus between each core and its corresponding L1 cache. Also, when high density z-axis bonding is used (e.g., when DBI is used), this z-axis bus can be wide and it can be defined wholly within the x-y cross-section of the core and its L1 cache, as further described below. By being wholly contained within this cross-section, the z-axis bus would not consume routing resources around the core and its L1 cache. Also, the speed and width of this bus allows the bus to have a very high throughput bandwidth, which perfectly complements the high speed of the L1 cache.

As shown in FIG. 12, the 3D processor **1200** defines an L2 cache for each core on the same die on which the core is defined. In some embodiments, each core can access the other core's L2 cache through z-axis connections established through the face-to-face bonding of the two IC dies. Also, due to the size of the L1 cache, the 3D process **1200** in some embodiments does not use an L3 cache.

In some embodiments, different components of a processor core of a multi-processor core are placed on different dies. FIG. 13 illustrates an example of a 3D processor **1300** that has different parts of a processor core on two face-to-face mounted dies **1305** and **1310**. In this example, the first die **1305** includes multiple pipeline **1390**, with each pipeline having an instruction fetch (IF) unit **1312**, an instruction decode unit **1314**, an execution unit **1316** and a write-back unit **1318**. The second die includes the instruction memory **1322** and data registers and memories **1324**.

As shown, the instruction memory **1322** on the second die overlaps with the IF units **1312** on the first die **1305**. Also, the data registers and memories **1324** on the second die overlap with the execution units **1316** and the write-back units. Numerous vertical signal paths are defined between overlapping core components by the z-axis connections between the top two interconnect layers of the dies **1305** and **1310**, and by (1) vias between interconnect layers of each

die **1305** or **1310**, and/or (2) interconnect segments on interconnect layers of each die.

Through vertical signal paths, each IF unit **1312** retrieves instructions from the instruction memory and provides the retrieved instructions to its instruction decode unit **1314**. This decode unit decodes each instruction that it receives and supplies the decoded instruction to its execution unit to execute. Through the vertical signal paths, each execution unit receives, from the data registers and memories **1324**, operands that it needs to execute a received instruction, and provides the result of its execution to its write-back unit **1318**. Through the vertical signal paths, each write-back unit **1318** stores the execution results in the data registers and memories **1324**. Other embodiments use other architectures to split a processor core between two different dies. For instance, some embodiments place the instruction decode and execution units **1314** and **1316** on different layers than the instruction fetch and write back units **1312** and **1318**. Still other embodiments use other arrangements to split a processor core between different dies. These or other embodiments put different ALUs, or different portions of the same ALU, of a processor core on different vertically stacked dies (e.g., on two dies that are face-to-face mounted through a DBI bonding process).

As mentioned above, it is advantageous to use DBI connections to connect overlapping connected regions on two dies that are vertically stacked because DBI allows for far greater density of connections than other z-axis connection schemes. FIG. 14 presents an example that illustrates this. This figure shows a compute circuit **1415** on a first die **1405** that overlaps a memory circuit **1420** on a second die **1410**, which is vertically stacked over the first die **1405**. The compute circuit can be any type of compute circuit (e.g., processor cores, processor pipeline compute units, neural network neurons, logic gates, adders, multipliers, etc.) and the memory circuit can be any type of memory circuit (e.g., SRAMs, DRAMs, non-volatile memories, caches, etc.).

In this example, both circuits **1415** and **1420** occupy a square region of 250 by 250 microns on their respective dies **1405** and **1410** (only the substrate surfaces of which are shown in FIG. 14). Also, in this example, a 100-bit z-axis bus **1425** is defined between these circuits, with the term bus in this example referring to the data and control signals exchanged between these two circuits **1415** and **1420** (in other examples, a bus might only include data signals). FIG. 14 illustrates that when TSVs are used to define this z-axis bus **1425**, this bus will consume on each die a region **1435** that is at least 2.5 times as large as the size of either circuit on that die. This is because TSVs have a 40 micron pitch. For the TSV connections, the two dies **1405** and **1410** will be front-to-back mounted with the TSVs going through the substrate of one of the two dies.

On the other hand, when the two dies are face-to-face bonded through mounted, and DBI connections are used to define the 100-bit z-axis bus **1425**, the cross-section **1430** of the DBI bus can be contained within the footprint (i.e., the substrate region) of both circuits **1415** and **1420** on their respective dies. Specifically, assuming that the DBI connections have a 2 micron pitch, the 100 DBI connections can be fit in as little as 20-by-20 micron square, as the 100 connections can be defined as a 10-by-10 array with each connection having a minimum center-to-center spacing of 2 microns with its neighboring connections. By being contained within the footprint of the circuits **1415** and **1420**, the DBI connections would typically not consume any precious routing space on the dies **1405** and **1410** beyond the portion already consumed by the circuits. In some embodiments,

DBI connections can have a pitch ranging from less than 1 micron (e.g., 0.2 or 0.5 microns) to 5 microns.

As the numbers of bits increase in the bus **1425**, the difference between the amount of space consumed by the TSV connections and the space consumed by DBI connections becomes even more pronounced. For instance, when 3600 bits are exchanged between the two circuits **1415** and **1420**, a 60-by-60 TSV array would require a minimum 2400-by-2400 micron region (at a 40-micron DBI pitch), while a 60-by-60 DBI array would require a minimum 120-by-120 region (at a 2-micron DBI pitch). In other words, the TSVs would have a footprint that is at least 400 times greater than the footprint of the DBI connections. It is quite common to have a large number of bits exchanged between a memory circuit and a compute circuit when performing computations (e.g., dot product computations) in certain compute environments (e.g., machine-trained neural networks). Moreover, the density of DBI connections allows for very large bandwidth (e.g., in the high gigabytes or in the terabytes range) between overlapping compute and memory circuits.

The density of DBI connection is also advantageous in connecting overlapping circuit regions on two dies that are vertically stacked. FIG. **15** presents an example that illustrates this. This figure shows two overlapping compute circuits **1515** and **1520** on two vertically stacked dies **1505** and **1510**. Each of the circuits occupies a 250-by-250 micron square on its corresponding die's substrate, and can be any type of compute circuit (e.g., processor cores, processor pipeline compute units, neural network neurons, logic gates, adders, multipliers, etc.).

Like the example in FIG. **14**, the example in FIG. **15** shows that when DBI connections are used (i.e., when the two dies **1505** and **1510** are face-to-face mounted through DBI), and the DBI connections have a 2-micron pitch, a 100-bit bus **1525** between the two circuits **1515** and **1520** can be contained in a region **1530** that is 20-by-20 micron square that can be wholly contained within the footprints of the circuits. On the other hand, when TSV connections are used (e.g., when the two dies are face-to-back mounted and connected using TSVs), the 100-bit bus **1525** would consume at a minimum a 400-by-400 micron square region **1535**, which is larger than the footprint of the compute circuits **1515** and **1520**. This larger footprint would consume additional routing space and would not be as beneficial as the smaller footprint that could be achieved by the DBI connections.

High density DBI connections can also be used to reduce the size of a circuit formed by numerous compute circuits and their associated memories. The DBI connections can also provide this smaller circuit with very high bandwidth between the compute circuits and their associated memories. FIG. **16** presents an example that illustrates these benefits. Specifically, it illustrates the reduction in the size of an array **1600** of compute circuits **1615** on a first die, by moving the memories **1620** for these circuits to a second die **1610** that is face-to-face mounted with the first die **1605** through DBI bonding process. In this example, a 6-by-10 array of compute circuits is illustrated, but in other examples, the array can have larger number of circuits (e.g., more than 100 circuits, more than 1000 circuits). Also, in other embodiments, the compute circuits and their associated memory circuits can be organized in an arrangement other than an array.

The compute circuits **1615** and the memory circuits **1620** can be any type of computational processing circuits and memory circuits. For instance, in some embodiments, the circuit array **1600** is part of an FPGA that has an array of

logic circuits (e.g., logic gates and/or look-up tables, LUTs) and an array of memory circuits, with each memory in the memory array corresponding to one logic circuit in the circuit array. In other embodiments, the compute circuits **1615** are neurons of a neural network or multiplier-accumulator (MAC) circuits of neurons. The memory circuits **1620** in these embodiments store the weights and/or the input/output data for the neurons or the MAC circuits. In still other embodiments, the compute circuits **1615** are processing circuits of a GPU, and the memory circuits store the input/output data from these processing circuits.

As shown in FIG. **17**, a memory array is typically interlaced with the circuit array in most single die implementations today. The combined length of the two interlaced arrays is X microns in the example in FIG. **17**. To connect two circuits in the same column in the array, the wiring would have to be at least X microns. But by moving the memory circuits onto the second die **1610**, as shown in FIG. **16**, two circuits in the same column can be connected with a minimum wiring length of X/2 microns.

Moreover, each memory circuit can have a higher density of storage cells as less space is consumed for defining shared peripheral channels for outputting signals to the circuits, as these output signals can now traverse in the z-axis. Also, by moving the memory circuits onto the second IC die **1610**, more routing space is available in the open channels **1650** (on the substrate and metal layers) between the compute circuits in the compute array **1600** on the first die **1605**, and between the memory circuits in the memory array **1602** on the second die **1610**. This additional routing space makes it easier to connect the outputs of the compute circuits. In many instances, this extra routing space allows these interconnects to have shorter wire lengths. It also makes it easier for compute circuits in some embodiments to read or write data from the memory circuits of other compute circuits. The DBI connections are also used in some embodiments to route signals through the metal layers of the second die **1610** in order to define the signal paths (i.e., the routes) for connecting the compute circuits **1615** that are defined on the first die **1605**.

The higher density of DBI connections also allow a higher number of z-axis connections to be defined between corresponding memory and compute circuits that are wholly contained within the footprints (i.e., within the substrate regions occupied by) of a pair of corresponding memory and compute circuits. As mentioned before, these DBI connections connect the top interconnect layer of one die with the top interconnect layer of the other die, while the rest of the connection between a pair of memory and compute circuits is established with interconnects and vias on these dies. Again, such an approach would be highly beneficial when the compute circuits need wide buses (e.g., 128 bit buses, 256 bit buses, 512 bit buses, 1000 bit buses, 4000 bit buses, etc.) to their corresponding memory circuits. One such example would be when the array of compute circuits are arrays of neurons that need to access a large amount of data from their corresponding memory circuits.

FIGS. **18** and **19** illustrates two examples that show how high density DBI connections can be used to reduce the size of an arrangement of compute circuit that is formed by several successive stages of circuits, each of which performs a computation that produces a result that is passed to another stage of circuits until a final stage of circuits is reached. In some embodiments, such an arrangement of compute circuits can be an adder tree, with each compute circuit in the tree being an adder. In other embodiments, the circuits in the

arrangement are multiply accumulate (MAC) circuits, such as those used in neural networks to compute dot products.

The examples in FIGS. 18 and 19 both illustrate one implementation of a circuit 1800 that performs a computation (e.g., an addition or multiplication) based on eight input values. In some embodiments, each input value is a multi-bit value (e.g., a thirty-two bit value). The circuit 1800 has three stages with the first stage 1802 having four compute circuits A-D, the second stage 1804 having two compute circuits E and F, and the third stage 1806 having a compute circuit G. Each compute circuit in the first stage 1802 performs an operation based on two input values. In the second stage 1804, the compute circuit E performs a computation based on the outputs of compute circuits A and B, while the compute circuit F performs a computation based on the outputs of compute circuits C and D. Lastly, the compute circuit G in the third stage 1806 performs a computation based on the outputs of compute circuits E and F.

FIG. 18 illustrates a prior art implementation of the circuit 1800 on one IC die 1805. In this implementation, the compute circuits A-G are arranged in one row in the following order: A, E, B, G, C, F and D. As shown, the first stage compute circuits A-D (1) receive their inputs from circuits (e.g., memory circuits or other circuits) that are above and below in the planar y-axis direction, and (2) provide their results to the compute circuit E or F. The compute circuits E and F provide the result of their computations to compute circuit G in the middle of the row. The signal path from the compute circuits E and F is relatively long and consumes nearby routing resources. The length and congestion of interconnects become worse as the size of the circuit arrangement (e.g., the adder or multiplication tree) grows. For instance, to implement an adder tree that adds 100 or 1000 input values, numerous adders are needed in numerous stages, which quickly results in long, big data buses to transport computation results between successive stages of adders.

FIG. 19 illustrates a novel implementation of the circuit 1800 that drastically reduces the size of the connections needed to supply the output of compute circuits E and F to the compute circuit G. As shown, this implementation defines the compute circuits A, B, E and G on a first die 1910, while defining the compute circuits C, D and F on a second die 1905 that is face-to-face mounted on the first die 1910 through a DBI bonding process. The compute circuits A, B, E, and G are defined in a region on the first die 1910 that overlaps with a region on the second die 1905 in which the compute circuits C, D and F are defined.

In this implementation, the compute circuit G is placed below the compute circuit E in the planar y-direction. At this location, the compute circuit G receives the output of the compute circuit E through a short data bus defined on the die 1910, while receiving the output of the compute circuit F through (1) z-axis DBI connections that connect overlapping locations 1950 and 1952 on the top interconnect layers of the dies 1905 and 1910, and (2) interconnects and vias on these dies that take the output of circuit F to the input of circuit G. In this implementation, the interconnects that provide the inputs to the compute circuit G are very short. The computation circuits E and G are next to each other and hence the signal path just includes a short length of the interconnect and vias between the circuit E and G. Also, the length of interconnects, vias, and z-axis DBI connections needed to provide the output of the compute circuit F to the compute circuit G is very small.

Hence, by breaking up the arrangement of the circuit 1800 between two dies 1905 and 1910, successive compute cir-

cuits can be placed closer to each other (because an additional dimension, i.e., the z-axis, is now available for placing circuits near each other), which, in turn, allows shorter interconnects to be defined between compute circuits in successive stages. Also, the high density of DBI connections makes it easier to define larger number of z-axis connections (that are needed for larger z-axis data buses) within the cross section of the regions that are used to define successive compute circuits.

Compute circuit arrangements can have more than three stages. For example, large adder or MAC trees can have many more stages (e.g., 8 stages, 10 stages, 12 stages, etc.). To implement such circuit arrangements, some embodiments (1) divide up the compute circuits into two or more groups that are then defined on two or more vertically stacked dies, and (2) arrange the different groups of circuits on these dies to minimize the length of interconnects needed to connect compute circuits in successive stages.

FIG. 20 presents an example to illustrate this point. This example shows one implementation of a compute circuit 2000 that performs a computation (e.g., an addition or multiplication) on sixteen multi-bit input values. This circuit includes two versions 2012 and 2014 of the compute circuit 1800 of FIGS. 18 and 19. The compute circuits in the second version are labeled as circuits H-N. Each of these versions has three stages. The outputs of these two versions are provided to a fourth stage compute circuit O that performs a computation based on these outputs, as shown.

To implement the four stage circuit 2000, the two versions 2012 and 2014 have an inverted layout. This is because the compute circuits A, B, and E (that operate on the first four inputs of the first version 2012) are defined on IC die 2010 while the compute circuits H, I and L (that operate on the first four inputs of the second version) are defined on the IC die 2005. Similarly, the compute circuits C, D, and F (that operate on the second four inputs of the first version 2012) are defined on IC die 2005 while the compute circuits J, K, and M (that operate on the second four inputs of the second version) are defined on the IC die 2010. Also, the third stage circuit G of the first version is defined on the IC die 2010, while the third stage circuit N is defined on the IC die 2005. The fourth stage aggregating circuit O is also defined on IC die 2010. Lastly, the second version 2014 is placed to the right of the first version in the x-axis direction.

This overall inverted arrangement of the second version 2014 with respect to the first version ensures that the length of the interconnect needed to provide the output of the third stage compute circuits G and N to the fourth stage compute circuit O is short. This is because, like compute circuits E, F and G, the compute circuits L, M, and N are placed in nearby and/or overlapping locations, which allows these three circuits L, M and N to be connected through short DBI connections, and mostly vertical signal paths facilitated by small planar interconnects plus several via connections. This arrangement also places compute circuits G, N and O in nearby and/or overlapping locations, which again allows them to be connected through short DBI connections, and mostly vertical signal paths facilitated by small planar interconnects plus several via connections.

FIG. 21 illustrates a device 2102 that uses a 3D IC 2100 (like any of the 3D IC 210, 200, 400, 600-900). In this example, the 3D IC 2100 is formed by two face-to-face mounted IC dies 2105 and 2110 that have numerous direct bonded connections 2115 between them. In other examples, the 3D IC 2100 includes three or more vertically stacked IC dies. As shown, the 3D IC die 2100 includes a cap 2150 that encapsulates the dies of this IC in a secure housing 2125. On

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the back side of the die **2110** one or more TSVs and/or interconnect layers **2106** are defined to connect the 3D IC to a ball grid array **2120** (e.g., a micro bump array) that allows this to be mounted on a printed circuit board **2130** of the device **2102**. The device **2102** includes other components (not shown). In some embodiments, examples of such components include one or more memory storages (e.g., semiconductor or disk storages), input/output interface circuit(s), one or more processors, etc.

In some embodiments, the first and second dies **2105** and **2110** are the first and second dies shown in any of the FIGS. **1-2**, **4**, **6-16**, and **19-20**. In some of these embodiments, the second die **2110** receives data signals through the ball grid array, and routes the received signals to I/O circuits on the first and second dies through interconnect lines on the interconnect layer and vias between the interconnect layers. When such data signals need to traverse to the first die, these signals traverse through z-axis connections crossing the face-to-face bonding layer.

FIG. **22** provides another example of a 3D chip **2200** that is formed by two face-to-face mounted IC dies **2205** and **2210** that are mounted on a ball grid array **2240**. In this example, the first and second dies **2205** and **2210** are face-to-face connected through direct bonded connections (e.g., DBI connections). As shown, several TSVs **2222** are defined through the second die **2210**. These TSVs electrically connect to interconnects/pads on the backside of the second die **2210**, on which multiple levels of interconnects are defined.

In some embodiments, the interconnects on the backside of the second die **2210** create the signal paths for defining one or more system level circuits for the 3D chip **2200** (i.e., for the circuits of the first and second dies **2205** and **2210**). Examples of system level circuits are power circuits, clock circuits, data I/O signals, test circuits, etc. In some embodiments, the circuit components that are part of the system level circuits (e.g., the power circuits, etc.) are defined on the front side of the second die **2210**. The circuit components can include active components (e.g., transistors, diodes, etc.), or passive/analog components (e.g., resistors, capacitors (e.g., decoupling capacitors), inductors, filters, etc.

In some embodiments, some or all of the wiring for interconnecting these circuit components to form the system level circuits are defined on interconnect layers on the backside of the second die **2210**. Using these backside interconnect layers to implement the system level circuits of the 3D chip **2200** frees up one or more interconnect layers on the front side of the second die **2210** to share other types of interconnect lines with the first die **2205**. The backside interconnect layers are also used to define some of the circuit components (e.g., decoupling capacitors, etc.) in some embodiments. As further described below, the backside of the second die **2210** in some embodiments can also connect to the front or back side of a third die.

In some embodiments, one or more of the layers on the backside of the second die **2210** are also used to mount this die to the ball grid array **2240**, which allows the 3D chip **2100** to mount on a printed circuit board. In some embodiments, the system circuitry receives some or all of the system level signals (e.g., power signals, clock signals, data I/O signals, test signals, etc.) through the ball grid array **2240** connected to the backside of the third die.

FIG. **23** illustrates a manufacturing process **2300** that some embodiments use to produce the 3D chip **2200** of FIG. **22**. This figure will be explained by reference to FIGS. **24-27**, which show two wafers **2405** and **2410** at different stages of the process. Once cut, the two wafers produce two

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stacked dies, such as dies **2205** and **2210**. Even though the process **2300** of FIG. **23** cuts the wafers into dies after the wafers have been mounted and processed, the manufacturing process of other embodiments performs the cutting operation at a different stage at least for one of the wafers. Specifically, some embodiments cut the first wafer **2405** into several first dies that are each mounted on the second wafer before the second wafer is cut into individual second dies.

As shown, the process **2300** starts (at **2305**) by defining components (e.g., transistors) on the substrates of the first and second wafers **2405** and **2410**, and defining multiple interconnect layers above each substrate to define interconnections that form micro-circuits (e.g., gates) on each die. To define these components and interconnects on each wafer, the process **2300** performs multiple IC fabrication operations (e.g., film deposition, patterning, doping, etc.) for each wafer in some embodiments. FIG. **24** illustrates the first and second wafers **2405** and **2410** after several fabrication operations that have defined components and interconnects on these wafers. As shown, the fabrication operations for the second wafer **2410** defines several TSVs **2412** that traverse the interconnect layers of the second wafer **2410** and penetrate a portion of this wafer's substrate **2416**.

After the first and second wafers have been processed to define their components and interconnects, the process **2300** face-to-face mounts (at **2310**) the first and second wafers **2205** and **2210** through a direct bonding process, such as a DBI process. FIG. **25** illustrates the first and second wafers **2405** and **2410** after they have been face-to-face mounted through a DBI process. As shown, this DBI process creates a number of direct bonded connections **2426** between the first and second wafers **2405** and **2410**.

Next, at **2315**, the process **2300** performs a thinning operation on the backside of the second wafer **2410** to remove a portion of this wafer's substrate layer. As shown in FIG. **26**, this thinning operation exposes the TSVs **2412** on the backside of the second wafer **2410**. After the thinning operation, the process **2300** defines (at **2320**) one or more interconnect layers **2430** the second wafer's backside. FIG. **27** illustrates the first and second wafers **2405** and **2410** after interconnect layers have been defined on the second wafer's backside.

These interconnect layers **2430** include one or more layers that allow the 3D chip stack to electrically connect to the ball grid array. In some embodiments, the interconnect lines/pads on the backside of the third wafer also produce one or more redistribution layers (RDL layers) that allow signals to be redistributed to different locations on the backside. The interconnect layers **2430** on the backside of the second die in some embodiments also create the signal paths for defining one or more system level circuits (e.g., power circuits, clock circuits, data I/O signals, test circuits, etc.) for the circuits of the first and second dies. In some embodiments, the system level circuits are defined by circuit components (e.g., transistors, etc.) that are defined on the front side of the second die. The process **2300** in some embodiments does not define interconnect layers on the backside of the second wafer to create the signal paths for the system level circuits, as it uses only the first and second dies' interconnect layers between their two faces for establishing the system level signal paths.

After defining the interconnect layers on the backside of the second wafer **2410**, the process cuts (at **2325**) the stacked wafers into individual chip stacks, with each chip stack include two stacked IC dies **2205** and **2210**. The process then mounts (at **2330**) each chip stack on a ball grid array and

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encapsulates the chip stack within one chip housing (e.g., by using a chip case). The process then ends.

In some embodiments, three or more IC dies are stacked to form a 3D chip. FIG. 28 illustrates an example of a 3D chip 2800 with three stacked IC dies 2805, 2810 and 2815. In this example, the first and second dies 2805 and 2810 are face-to-face connected through direct bonded connections (e.g., DBI connections), while the third and second dies 2815 and 2810 are face-to-back connected (e.g., the face of the third die 2815 is mounted on the back of the second die 2810). In some embodiments, the first and second dies 2805 and 2810 are the first and second dies shown in any of the FIGS. 1-2, 4, 6-16, and 19-20.

In FIG. 28, several TSVs 2822 are defined through the second die 2810. These TSVs electrically connect to interconnects/pads on the backside of the second die 2810, which connect to interconnects/pads on the top interconnect layer of the third die 2815. The third die 2815 also has a number of TSVs that connect signals on the front side of this die to interconnects/pads on this die's backside. Through interconnects/pads, the third die's backside connects to a ball grid array 2840 that allows the 3D chip 2800 to mount on a printed circuit board.

In some embodiments, the third die 2815 includes system circuitry, such as power circuits, clock circuits, data I/O circuits, test circuits, etc. The system circuitry of the third die 2815 in some embodiments supplies system level signals (e.g., power signals, clock signals, data I/O signals, test signals, etc.) to the circuits of the first and second dies 2805 and 2810. In some embodiments, the system circuitry receives some or all of the system level signals through the ball grid array 2840 connected to the backside of the third die.

FIG. 29 illustrates another example of a 3D chip 2900 with more than two stacked IC dies. In this example, the 3D chip 2900 has four IC dies 2905, 2910, 2915 and 2920. In this example, the first and second dies 2905 and 2910 are face-to-face connected through direct bonded connections (e.g., DBI connections), while the third and second dies 2915 and 2910 are face-to-back connected (e.g., the face of the third die 2915 is mounted on the back of the second die 2910) and the fourth and third dies 2920 and 2915 are face-to-back connected (e.g., the face of the fourth die 2920 is mounted on the back of the third die 2915). In some embodiments, the first and second dies 2905 and 2910 are the first and second dies shown in any of the FIGS. 1-2, 4, 6-16, and 19-20.

In FIG. 29, several TSVs 2922 are defined through the second, third and fourth die 2910, 2915 and 2920. These TSVs electrically connect to interconnects/pads on the backside of these dies, which connect to interconnects/pads on the top interconnect layer of the die below or the interconnect layer below. Through interconnects/pads and TSVs, the signals from outside of the chip are received from the ball grid array 2940.

Other embodiments use other 3D chip stacking architectures. For instance, instead of face-to-back mounting the fourth and third dies 2920 and 2915 in FIG. 29, the 3D chip stack of another embodiment has these two dies face-to-face mounted, and the second and third dies 2910 and 2915 back-to-back mounted. This arrangement would have the third and fourth dies 2915 and 2920 share a more tightly arranged set of interconnect layers on their front sides.

While the invention has been described with reference to numerous specific details, one of ordinary skill in the art will recognize that the invention can be embodied in other specific forms without departing from the spirit of the

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invention. For instance, in the examples illustrated in FIGS. 1-2, 4, 6-16, and 19-20, a first IC die is shown to be face-to-face mounted with a second IC die. In other embodiments, the first IC die is face-to-face mounted with a passive interposer that electrically connects the die to circuits outside of the 3D chip or to other dies that are face-to-face mounted or back-to-face mounted on the interposer. Some embodiments place a passive interposer between two faces of two dies. Some embodiments use an interposer to allow a smaller die to connect to a bigger die.

Also, the 3D circuits and ICs of some embodiments have been described by reference to several 3D structures with vertically aligned IC dies. However, other embodiments are implemented with a myriad of other 3D structures. For example, in some embodiments, the 3D circuits are formed with multiple smaller dies placed on a larger die or wafer. FIG. 30 illustrates one such example. Specifically, it illustrates a 3D chip 3000 that is formed by face-to-face mounting three smaller dies 3010a-c on a larger die 3005. All four dies are housed in one chip 3000 by having one side of this chip encapsulated by a cap 3020, and the other side mounted on a micro-bump array 3025, which connects to a board 3030 of a device 1935. Some embodiments are implemented in a 3D structure that is formed by vertically stacking two sets of vertically stacked multi-die structures.

The invention claimed is:

1. A three dimensional (3D) processor circuit, comprising:
 - a first integrated circuit (IC) die comprising a first processor core; and
 - a second IC die vertically mounted on the first IC die, the second IC die comprising a first cache for the first processor core,
 wherein the first cache vertically overlaps with at least a portion of the first processor core at an overlapping area, and
 - wherein the first IC die is communicatively coupled with the second IC die through a hybrid bond comprising a plurality of directly bonded metal contact pads and directly bonded non-conductive regions, and
 - wherein the first processor core is electrically connected to the first cache through at least one of the plurality of directly bonded metal contact pads disposed within the overlapping area.
2. The 3D processor circuit of claim 1, further comprising a plurality of z-axis connections between the first processor core and the first cache.
3. The 3D processor circuit of claim 2, wherein the plurality of z-axis connections comprises a z-axis bus, the z-axis bus being wholly defined within the overlapping area.
4. The 3D processor circuit of claim 2, wherein at least two of the plurality of z-axis connections have a center-to-center pitch that is less than 5 microns.
5. The 3D processor circuit of claim 2, wherein the plurality of z-axis connections comprises interconnect layers interconnecting the first processor core and the first cache through the hybrid bond.
6. The 3D processor circuit of claim 1, wherein the first IC die and the second IC die are face-to-face mounted.
7. The 3D processor circuit of claim 1, wherein:
 - the second IC die further comprises a second processor core;
 - the first IC die further comprises a second cache;
 - the second cache is for the second processor core; and
 - the second processor core vertically overlaps with the second cache.
8. The 3D processor circuit of claim 1, wherein the first cache is an L1 cache.

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9. The 3D processor circuit of claim 8, wherein the first cache completely overlaps the first processor core.

10. The 3D processor circuit of claim 8, further comprising a second cache on the first IC die for the first processor core.

11. The 3D processor circuit of claim 10, wherein the second cache is an L2 cache.

12. The 3D processor circuit of claim 1, wherein the 3D processor circuit excludes an L3 cache.

13. The 3D processor circuit of claim 1, wherein the 3D processor circuit is a graphical processing unit.

14. The 3D processor circuit of claim 1, wherein the 3D processor circuit is a central processing unit.

15. An electronic device, comprising: a three-dimensional (3D) processor circuit, comprising:

a first integrated circuit (IC) die comprising a first processor core;

a second IC die vertically mounted on the first IC die, the second IC die comprising a first cache for the first processor core,

wherein the first IC die is communicatively coupled with the second IC die through a hybrid bond comprising a plurality of directly bonded metal contact pads and directly bonded non-conductive regions,

wherein the first cache vertically overlaps with at least a portion of the first processor core at an overlapping area, and

wherein the first processor core is electrically connected to the first cache through at least one of the plurality of directly bonded metal contact pads disposed within the overlapping area; and

a board on which the 3D processor circuit is mounted.

16. The electronic device of claim 15, further comprising a plurality of z-axis connections between the first processor core and the first cache.

17. The electronic device of claim 16, wherein the plurality of z-axis connections comprises a z-axis bus, the z-axis bus being wholly defined within the overlapping area.

18. The 3D processor circuit of claim 16, wherein the plurality of z-axis connections comprises interconnect layers interconnecting the first processor core and the first cache through the hybrid bond.

19. The electronic device of claim 15, wherein the first IC die and the second IC die are face-to-face mounted.

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20. The electronic device of claim 15, wherein: the second IC die further comprises a second processor core;

the first IC die further comprises a second cache;

the second cache is for the second processor core; and

the second processor core vertically overlaps with at least a portion of the second cache.

21. A three dimensional (3D) processor circuit, comprising:

a first integrated circuit (IC) die comprising a processor core;

a second IC die vertically stacked on the first IC die, the second IC die comprising a cache for the processor core; and

interconnect layers interconnecting the first IC die and the second IC die through a hybrid bond comprising a plurality of directly bonded metal contact pads and directly bonded non-conductive regions,

wherein the processor core is electrically connected to the cache through at least one of the plurality of directly bonded metal contact pads disposed within an area defined by overlapping circuit blocks on the first and second IC dies.

22. The 3D processor circuit of claim 21, wherein the cache is a first cache and the first IC die further comprises a second cache.

23. The 3D processor circuit of claim 21, wherein adjacent metal contact pads have a pitch between 0.5 μm and 15 μm .

24. The 3D processor circuit of claim 21, wherein the first IC die and the second IC die are face-to-back mounted.

25. The 3D processor circuit of claim 21, wherein:

at least one of the first IC die and the second IC die comprises through silicon vias (TSVs); and

a signal path communicatively coupling the processor core and the cache further comprises at least one of the TSVs.

26. The 3D processor circuit of claim 21, wherein the overlapping circuit blocks comprise a first circuit block comprising at least a portion of the processor core and a second circuit block comprising at least a portion of the cache.

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